Phylum Annelida (Segmented worms)

- They have segmented body with structures and organs repeated along the body.
- They are **triploblastic. (Ectoderm, mesoderm, endoderm).**
- Have complete digestive system and well developed ganglionic nervous system.
- Have closed circulatory system.
- They have structure which is called **chaetae** or **parapodia** projections found on every segment for locomotion.

Note: ***chaetae*** are tiny hair like structures.

- They are free living which are found in moist soil and water.
- Important to increase soil fertility and medicinal Leech

Examples: Earth worm, Leech

Phylum Mollusca

- They have unsegmented spotted bodies which are covered by **shells.** Some are shell less.
- Their bodies are divided in to head, foot and visceral mass.
- They are **triploblastic. (Ectoderm, mesoderm, endoderm)**
- They are coelomates (ie. Have true body cavity) consisting of reproductive structure, kidney and heart.
- They breathe by using gills.
- Some have tentacles example: Octopus
- They are the most intelligent of all invertebrates and have wide range of life styles.

Examples: snail, squids and octopus

Figure 5.5: Octopus

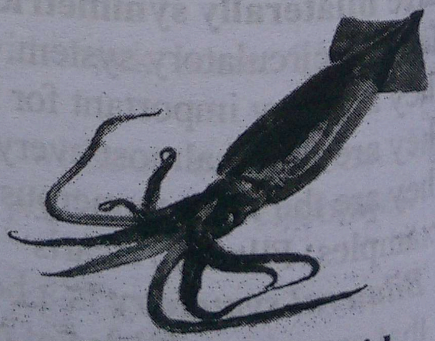

Figure 5.6: Squid

Phylum Echino dermata

- They are spiny skinned animals.
- They are triploblastic.
- Have mouth, gut and anus.
- They have tube feet for locomotion and obtaining food.
- All are marine animals.

Examples: Asteris, sea urchins, starfish, paracucumana tricolour which are known as called **sea apples.**

Phylum - Platyhelminthes (Flatworms)

- They have relatively high level of organization. Thus they illustrate organ system level of organization.
- Have flattened body with mouth but no anus.
- They are **acoelomate** (i.e. no internal body cavity).
- They are **triploblastic** (three layered body). These are:
  - Ectoderm
  - Mesoderm
  - Endoderm
- They use diffusion for material transport
- They are hermophrodites.
- They live as **parasites** or **Free living**
- **Examples:** Planaria, Flukes, Tapeworm

Figure: 5.7 Planaria

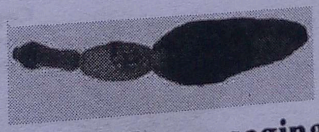

Figure 5.8: Tanea saginata (Tape worm)

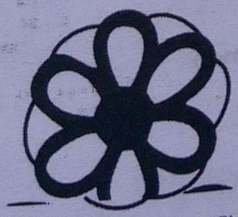

Figure 5.9: Sea – urchin

Phylum Arthropoda

- They have jointed Feet (limbs).
- They are the most varied animals consisting of various species.
- Have external exoskeleton which is composed of chitin.
- Have segmented body.
- Consists of body divided in to head, thorax and abdomen.
- Have well developed nervous system and a complete gut from mouth to anus.
- Have an open circulatory system.
- Have body cavity (coelmate).
- Consists of five (5) major classes based on number of limbs, presence and absence of antennae and number of body parts.
  - Class Arachinida
  - Class crustecea
  - Class Insecta
  - Class chilopoda
  - Class Diplopoda, some of them will be discussed.

Class Archinida

- They have body divided in to **cephalothorax** and **abdomen**.
- They have four pair of legs and no antenna.
- The gas exchange is by **tracheal system**.
- Some are poisonous.
- Some have another structure for exchange of gas called **lamellae,** also called **book lung**. It is a unique feature for arachnoids.

Examples: spider, scorpion, Ticks and mites

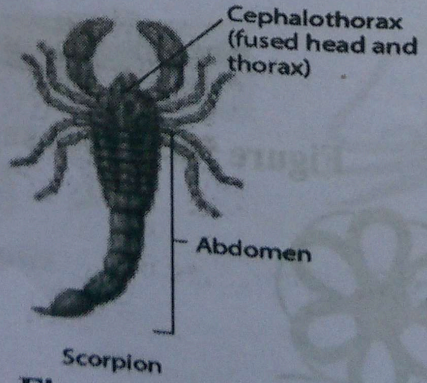

Figure 5.10: Scorpion

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Class Insect

- The numerous groups of all arthropods.
- They have body divided in to head, thorax and abdomen.
- Consists of a pair of antennae, three pairs of legs and one or two pairs of wings and compound eyes.
- They respire by **trachea** and excrete by means of **malphigian tubules.**
- They live almost everywhere but mostly land based.

Examples: flies, butter Flies, Moth, beetles, wasp, bees etc.

Class crustacean

- They have two body parts, **cephalothoraxes** (head fused with thorax) and **abdomen**.
- They have four pairs of jointed legs.
  - Two pairs of antennae (Unique to crustaceans) and simple eyes.
- Their body is protected by structure called **carapace.**

Examples: shrimp, lobsters, crab, Daphnia, water flea, Cray fish etc,

Class chilopoda

- Have long and flattened body
- Consists of a pair of antennae, pair of jaws, pair of legs per segment.
- They use tracheal system for respiratory purpose and excrete by malphigian tubules. Examples: Centipedes.

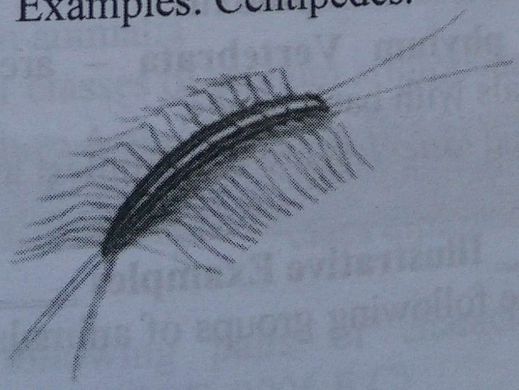

Figure 5.11: Centipedes

Class Diplopada

- Have long and cylindrical body.
- Has a pair of segmented antennae and two pair of legs per segment.
- Tracheal system is used for respiration and malphigian tubules for excretion.

Examples: Millipedes

Figure 5.12: Millipedes

Phylum Chordata

- Consists of animals which have **notochord** at the early stage of their life cycle.

  Note: ***Notochord*** – is a Flexible rod like structure of cartilage running along side of the body.
- They have gill slits at the embryonic stage which is later replaced by lungs and gills.
- They have a hallow nerve chord.
- Phylum chordata consists of the following subphylas:
  - **Sub phylum urochordata** – consists of marine animals such as: **Tunicate**
  - **Sub phylum cephalochoradata** – includes animals like **lancelets** and **amphioxus.**
  - **Sub phylum Vertebrata** – are animals that include animals with back bones.

Note: ***Tunicates*** and ***lancelets*** are the simplest and most primitive forms of chordates.

Illustrative Examples

35. Which of the following groups of animals has the largest number of species?
    A. Mollusea
    B. Arachnid
    C. Vertebrates
    D. Insect

Explanation: Class insecta which is located in phylum Arthropoda are the most numerous groups of organisms on the earth.

Answer: D

Question number 36 is based on the following groups of organisms

1. Flat worm
2. Arthropods
3. Fish
4. Birds
5. Repitiles
6. Amphibian
7. Molluscus

36. Animals with back bone include:
   A. 1,2,3, and 4
   B. 3,4,5 and 6
   C. 4,5,6 and 7
   D. 1,3,5 and 7

Explanation:
Among vertebrates (animals with back bone) the most known classes are; class pisces, class Amplibia, class repitila, Class aves and class mammalia. Representatives of each class are respectively, fish, amphibians, repitiles, birds and mammals.

Answer: B

37. Which of the following in **NOT** a member of the phylum Arthropoda?
   A. Arachinida
   B. Crustacea
   C. Insecta
   D. Coclenterata

Explanation:
Phylum Arthropoda, consists of the following classes, Arachinida, crustecea, insecta, chilopoda, diplopoda.

Answer: D

38. The most numerous organisms in terms of species in kingdom of animalia are the (EGSECE 2003)
   A. Arthropods
   B. Nematodes
   C. Molluscs
   D. Chordates

Explanation: Arthropods are the most numerous groups of all phylas of kingdom animalia

Answer: A

39. List out the major classes of phylum arthropoda.

Explanation: Phylum Arthropoda has five classes. These are:

- Class arachnida
- Class crustecea ➢ class chilopoda
- Class Insecta ➢ class diplopoda

40. Which of the following pairs of characteristics described kingdom Monera? (EGSECE, 2006 E.C)
   A. Multicellular and prokaryotic
   B. Multicellular and eukaryotic

C. Single – celled and prokaryotic
D. Single – celled and eukaryotic

Explanation: Kingdom Monera consists of groups of organisms which are characterized by the following major characteristics:

- They are prokaryotic and unicellular.
- They have either autotrophic or heterotrophic mode of nutrition.
- They reproduce asexually by fragmentation and binary fusion.

Examples: Bacteria and blue – green algae.

Answer: C

41. This item is based on the following list of groups of animals. (EGSECE, 2006 E.C)

1.	Sponges	4.	Amphibians
2.	Pisces	5.	Reptiles
3.	Nematodes	6.	Annelids

42. Which of the following groups of animals are classified under phylum chordata?

A. 2,3,6 C. 1,5,6
B. 1,2,5 D. 2,4,5

Explanation: phylum chordata consists of the following major classes:

- Class pisces (Fish)
- Class amphibian (amphibians)
- Class repitalia (reptiles)
- Class aves (birds)
- Class mammalia (mammals)

Answer: D

Post Test 4

Part I: Choose the best answer from the given alternatives.

1. Which one of the following is NOT the characteristic feature of kingdom animalia?
   A. Their body has complex organ system.
   B. They have chloroplasts and cell wall.
   C. They reproduce sexually and some reproduce asexually.
   D. They exhibit locomotion.
2. Animals that don't posses backbone are:
   A. Vertebrates C. Chordates
   B. Invertebrates D. Hermaphrodites

3. Which
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   B.
   C.
   D.

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Which of the following class of animals in NOT invertebrate?
A. Aves B. Insecta C. Arachinida D. Crustecea

Animals with notochord are found in a phylum:
A. Arthropoda C. Mollusca
B. Chordata D. Platyhelmentes

Arthropods are characterized by all of the following except:
A. have jointed Feet (limbs)
B. have external exoskeleton which is composed of chitin
C. have unsegmented body
D. Consists of body divided in to head, thorax and abdomen.

Vertebrates

Sub phylum – vertebrata consists of animals that have back bone
Have the following common features:

- Closed circulatory system.
- Two pair of limbs.
- Excretory organs such as kidney.
- Internal skeleton (endoskeleton).

Subphylum vertebrata is divided in to the Following classes:

- Class Pisces – Fish
- Class Amphibia – Amphibians (Frog, salamander and Toad)
- Class Reptilia – reptiles (snakes, Lizards, Turtles, Tortoises and crocodiles)
- Class Aves – Birds
- Class Mammalia – Mammals

Class Pisces

Have stream lined bodies with scaly skin.
Have four fins for swimming.
They live in water except mudskipper and lung Fish.
Their breathing organ is **gill**.
Their heart is two chambered, (Auricle and Ventricle.)
Have external fertilization.
They are piokilothermic animals.

Note: ***Poikilothermic*** animals have variable body temperature that changes with environmental temperature.

- They use their lateral line system which is used for hearing
- Fishes can be grouped in to two subclasses. These are

Bony Fish and cartilaginous Fish

Comparison between bony fish and cartilaginous fish

Bony fish (teleosts)	Cartilaginous fish
Have bony skeleton	Have cartilaginous skeleton
Have round – shaped scales	Have scales that are not round shaped
Have opercula (gill covers) covering their gills	Have no opercula (gill covers) but have gill slits.
Have homocercal tails (even sized fins)	Have heterocercal tails (one part is larger than the other)
Are usually smaller in size	Are usually larger in size
Example: Tilapa, Nile perch, cod, mackerel, cat fish.	Examples: Sharks, skates, rays.

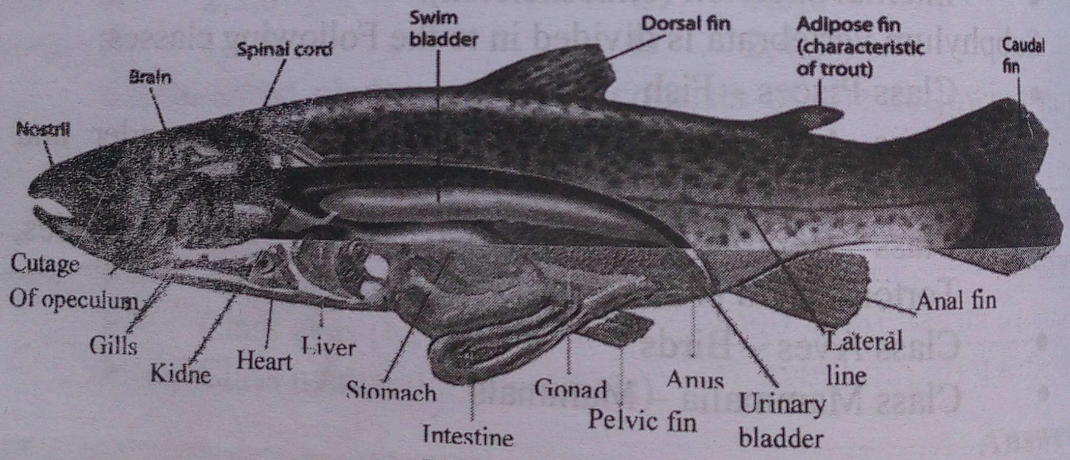

Figure 5.13: Fish

Class Amphibia

The word **amphibian** is derived from two Greek words, **amphi** means **both** and **bios** means **life**. This is because amphibians spend parts of their lives (larvae or tadpoles) in water and adults on the land.

- They are the first vertebrates to colonize the land.
- They have simple sac – like lungs which are less efficient and moist skin for breathing.

- Their reproduction involves external fertilization which needs water as a fertilizing agent. They spend their larval stage in aquatic environment.
- They are ectothermic (piokilothermic). Thus, they depend on external temperature to regulate their body temperature.
- Amphibians are on the threat of extinction because of:
  - Pathogenic fungus that largely kills amphibians.
  - Loss of habitat
  - Pollution
- Their lifecycle includes metamorphosis which includes stages from larva to adult.

Examples: Frog, Toad, Salamanders

Note: Amphibians are vertebrates that spend part of their lives in water and part of it on land.

Comparison of Frog and toad

Frog	*Toad*
Has a smooth skin	Has rough skin
Has moist ski	Has dry skin
Has more webbed feet	Has less webbed feet
Has brightly colored body	Has cell colored body
Has more streamlined body	Has less stream lined body
Has extra long hind legs	Has hind legs that are not extra long

Class Aves

- Have skin that bears Feathers and scales on legs.
- Have bony skeleton
- Two pair of Pentadectyl limbs, front pair forms wings.

Note: Pentadectyl limbs **are five digit limbs.**

- Visceral clefts never develop gills.
- Have no external ear.
- They have internal fertilization

Note: **Internal Fertilization** takes place inside the female body.

- They are homeothermic (endothermic) animals, thus they have constant body temperature which is not affected by changes in the environmental temperature.
- Examples are pegion, agles.

Class Repitilia

- Consists of all reptiles.
- Have dark scaly skin with horny scales.
- Have bony skeleton.
- Two pair of pentadectyl limbs usually present
- Visceral clefts never develop gills. They use lungs for gas exchange.
- Have no external ear.
- They use internal fertilization. They lay fertilized egg on land. Their eggs have leathery skin
- They are piokilothermic (ectothermic).

Examples: snake, crocodile, Lizards, aligators, Turtles tortoises.

- Dinosaurs were reptiles.

Class Mammalia

- Their skin bears hairs. The skin consists of glands, like sebaceous and sweat.
- They have bony skeleton.
- Have two pairs of pentadectyl limbs.
- Visceral clefts never develop gills. They use lungs for gas exchange.
- Have external ear (in addition to middle and inner ears).
- They are homoeothermic (endothermic)
- Class mammalia has three groups, these are:
  - Egg lying mammals
  - Marsupials
  - Placental mammals

Egg lying Mammals

- They are primitive groups.

- They lay eggs.
- Have webbed Feet.
- **Examples**: Duckbilled platypus

Marsupials

- They produce tiny immature young which remain in marsupium (pouch) to complete their development.
- They get milk from mammary glands.

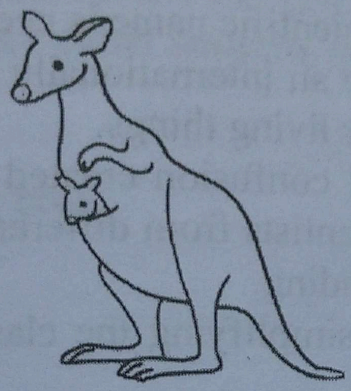

Fig. 5.14: Kangaroo (marsupial mammal)

Placental mammals

Placental mammals have complex placenta in which exchange of material takes place between the mother and the foetus, until the embryonic development is completed.

- Their body is nourished by milk from the mother which is produced by mammary gland.
- These are also called **higher Mammals.**

Examples include: cows, dogs, cats, elephants, lions, humans, etc.

Illustrative Example

42. Why mammals are considered as the most successful animals? Because mammals (EGSECE 2004E.C)
    A. Reproduce by sexual reproduction
    B. Are adapted to aquatic environment
    C. Regulate their body temperature
    D. Are the first vertebrate to colonize land

Explanation: Mammals are adapted to live in different habitats.

Answer: B

44. What major problem did the scientific naming of organisms solve? (EGSECE, 2005 E.C)
A. Showed the mechanism how new species arise from the existing ones.
B. Provided understanding among scientists of different countries.
C. Designed the hierarchy of the different groups of living things.
D. Showed the way how living organisms originated on this planet

Explanation: Scientific name is needed for various purposes:
- To create an internationally accepted way of referring to a particular living things.
- To avoid confusion created by different languages which helps scientists from different countries to have a common understanding.
- To help simplifying the classification and study of living things.

Answer: B

45. Which of the following is true about kingdom animalia? (EGSECE, 2005 E.C)
A. They have cell wall
B. They are immobile
C. They are autotrophs
D. They have nervous system

Explanation:
- To common features of organisms in kingdom animals is:
  - They are heterotrophs
  - They have nervous system
  - They are multicellular and eukaryotic
  - They exhibit locomotion.

Answer: D

46. Which of the following organisms is vertebrate animal? (EGSECE, 2005 E.C)
A. Lizard
B. Ascaris
C. Earth worm
D. snail

Explanation:
- **Vertebrates** are animals with back bone. They are all included in phylum choradata. Phylum chordata has the following major classes:
  - Class Pisces eg. Fish
  - Class repitilia – eg. Lizard

- Class aves – eg. Birds
- Class amphibian – eg. Frogs
- Class mammalian – eg. Human being

45. Describe the common features of all animals. **Answer: A**

Explanation:

All animals share the following common features:

- They are multicellular, eukaryotic and heterotrophic.
- They have locomotory structures.
- They posses no chloroplasts and cell wall.
- They can reproduce either sexually or asexually.

46. Explain the meaning of:

- Hermaphrodites
- Radial symmetric body
- Vertebrates
- Invertebrates
- Chaetae
- endoderm
- ectoderm

Explanation:

- **Hermaphrodite** – an organism having both male and female sex organs.
- **Radically symmetric body** – a body which can be divided in to many halves which are mirror image of one another.
- **Bilaterally symmetric body** – is a body which is divided in to two halves that are mirror images of one another.
- **Vertebrates** – are animals with back bone.
- **Invertebrates** – are animals without back bone.
- **Chaetae** – the locomotry structures of annelids.
- **Endoderm** – animals that have constant body temperature which is not affected by external environment.
- **Ectoderm** – Animals which have variable body temperature.

47. What are the criteria to group arthropods in to different classes?

Explanation: Arthropods have joined limbs and they are classified in to different classes based on:

- ✓ The number of their limbs.
- ✓ Presence of absence of antennae
- ✓ Parts of their body

49. Compare and contrast:
    i) Class insecta and class crustacean
    ii) Class mammalian and Pisces

Explanation:

i) Comparison between class insect and class crustacean

Class insect	Class crustecea
• Have body divided in to head, thorax and abdomen • Have three pairs of legs. • Have a pair of antennae Examples: flies, moth, beetles.	• Have body divided into head and cephalothorax. • Have four pairs of legs • Have two pairs of antennae Example: shrimp, crab, water flea

50. List out the common features of all chordates.

Explanation:

All chordates share the following common characteristics.

- Have notochord at early stage of their life.
- Have gill slits at their embryonic stage.
- Have hallow nerve chord.

51. List out the major classes of subphylum vertebrata.

Explanation:

- Sub phylum vertebrata consists of the following major classes:
  - Class pisces
  - Class amphibian
  - Class Repitalia
  - Class Aves
  - Class Mammalia

52. What makes amphibians so different from other vertebrates?

Explanation:

- **Amphibians** differ from other vertebrates because they spent part of their lives in water and on the land.

53. Distinguish between:
    - External fertilization and internal fertilization

Explanation:

- **External fertilization** is the fusion of the male and female gamete outside the body of the female reproductive organ
- **Internal fertilization** is the union of the male and female gamete inside the female reproductive organ.

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54. What is pentadectayl limb?

Explanation: Pentadectyl limb is a five digit limb which serves as an evidence in the study of evolution and taxonomy.

55. Describe the unique features of:

- Fishes (Pisces)
- Birds (aves)
- Reptiles
- Mammals
- Amphibians

Explanation:

Animal	Unique feature
Fish	• Have fins for swimming • Have gills for breathing
Birds	• Have feathery skin
Reptiles	• Have dark scaly skins
Mammals	• Have body converged with hair • nourish milk to their young
Amphibians	• Have ability to spend some part of their life cycle in water and land.

56. Compare between class mammalian and class Pisces.

Explanation:

Mammals	Pisces
• Have body which is covered with hair. • Their breathing organ is lung. • They have four chambered heart • They have internal fertilization • They are endothermic (homoeothermic)	• Have stream lined body with scaly skin. • Their breathing organ is gill • They have two chambered heart • They have external fertilization • They are ectothermic (piokilothermic)

57. List out the three groups of class mammalian and give examples for each.

Explanation: Organisms of class mammalian can be grouped as:

- Egg lying mammals – example: Duckbilled platypus
- Marsupials – example: kangaroo

- Placental mammals – example: Cows, dogs, cats, elephant, etc.

58. Living things are classified in order to: (EGSECE, 2006 E.C)
A. Make things sound complicated
B. Group various dissimilar organisms in to one.
C. Bring organisms in to chaotic environment.
D. Understand how life is originated.

Explanation: Living things are classified to:
- place them in groups based on their similarities and differences.
- Study the relationship between organisms in order to understand how they are originated.
- Simplify the study of living organisms.
- Bring order out of chaos and confusions.

Answer: D

59. Which of the following organism is a fungus?
A. *Entamoeba hytolyca*
B. *Candida albicans*
C. *Plasmodium placifarium*
D. *Mycobacterium tuberculosis*

Explanation: *Candida albicans* is a fungial species that cause a disease called thrush.

Entamoeba hystolica	Protozoa
Plasmodium palcifarium	Protozoa
Mycobacterium tuberculosis	Bacteria

Answer: B

60. Which of the following divisions of plant is correctly matched with their representative example?
A. Angiospermae ⟶ *Zea mays*
B. Pteridophyta ⟶ *punus sylvestris*
C. Bryophyte ⟶ *Dryopter is concolor*
D. Gymnospermae ⟶ *Lantana camara*

Explanation: *Zea mays* is a scientific name for maize which is an angiosperm (monocot) plant.

Answer: A

Post Test 5

Part I: Choose the best option from the given alternatives.

1. An organism that posses both male and female sex organs is:
A. Marsupial
B. Hermaphrodite
C. sponges
D. mammals

2. Which if the following is a marsupial animal?
   A. Duck C. Cow
   B. Kangaroo D. Bat
3. Which one of the following is the unique feature of mammals?
   A. Feathery skin C. Gill as a breathing organ
   B. Scaly skin D. Hairy skin
4. All of the following are placental animals except one:
   A. Cow C. Dog
   B. Kangaroo D. Elephant
5. Which one of the following animal belongs to class repitila?
   A. Cat B. fish C. Crocodile D. Frog
6. Having features such as external fertilization, scaly skin, gills belongs to class:
   A. Pisces B. Reptiles C. mammalian D. Amphibian
7. Class of vertebrates that spend part of their lives in water and on the land are:
   A. Amphibians B. Aves C. mammals D. Reptiles
8. Feathery skin is the characteristic feature of:
   A. Birds B. Pisces C. Reptiles D. Mammals

Supplementary Exercise

1. In 'five – kingdom' system of classification, organisms are classified based on the similarity of:
   A. Analogous structures
   B. Homologous structures
   C. Artificial breeding
   D. The functions of body parts
2. Groups of organisms that can inter breed successfully with one another to produce fertile offspring are grouped under the same:
   A. Kingdom C. species
   B. Genus D. Phylum
3. The branch of biology which is concerned with classification of organisms is:
   A. Physiology C. Morphology
   B. Evolution D. Taxonomy
4. Which of the following pathway is correct with increasing its complexity
   A. Genus → order → phylum → kingdom
   B. kingdom → Genus → order → phylum
   C. phylum → kingdom → Genus → order
   D. order → phylum Genus → kingdom

5. Which of the following is correct about ***Homo sapiens?***
   A. It's a common name
   B. Homo is a genus name
   C. Sapiens is a phylum name
   D. It's an **English** word for naming humans.
6. Which of the following organisms are excluded from the five kingdom system of classification?
   A. Algae
   B. Fungi
   C. Viruses
   D. Bacteria's
7. Bacteria's differ from organisms of other kingdoms by:
   A. Mode of nutrition
   B. Asexual reproduction
   C. Prokaryotic cellular nature
   D. having cell membrane
8. Which of the following is incorrect about protozoan?
   A. They are microscopic
   B. They have chloroplasts
   C. They have locomotory structures
   D. They are either parasitic or Free-living
9. Which of the following phyla of sub kingdom protozoa are is correctly matched with their examples.
   A. Phylum sarcomastigophora – **Fiardia**
   B. Phylum ciliophora – plasmodium
   C. Phylum Apicomplexa – plasmodium
   D. Phylum ciliophora – paramecium
10. The snail is classified in the group, (ESLCE – 1989 E.C)
    A. Amphibians
    B. Echinoderms
    C. Arthropods
    D. Mollusks
11. Which major group of the plant kingdom is best characterized by having needle shaped leaves and production of seeds in structures called cones?
    A. Ferns
    B. Conifers
    C. Monocotyledons
    D. Angiosperms
12. Nearly all single celled eukaryotes that are either heterotrophic or photosynthetic belong to the kingdom (EGSECE 1997 E.C)
    A. Fungi
    B. Animalia
    C. Plantae
    D. Protista

13. Monocotyledons differ from dicotyleadons by having:
    A. Parallel leaf venation
    B. Tap root system,
    C. Presence of vascular cambium
    D. Two cotyledon
14. Which of the following is least related to the others?
    A. Teff
    B. Maize
    C. Conifer
    D. Rose
15. Which of the following organisms are expected to be more related?
    A. *Felis domesticus* and *Homo sapiens*
    B. *Hagenea abysinica* and *panthera leo*
    C. *Panthera pardus* and *panthera leo*
    D. *Phaseolus vulgans* and *Zea mays*
16. The common characteristics of Fishes, amphibians, reptiles, birds and mammals is (ESLCE – 1979 E.C)
    A. Variable body temperature
    B. Egg lying
    C. Constant body temperature
    D. Presence of back bone
17. Which of the following groups of animals has the largest number of species?
    A. Mollusca
    B. Arachinia
    C. Vertebrates
    D. Insect

Question number 18 is based on the following groups of organisms:

1. Flat worms
2. Arthropods
3. Fish
4. Birds
5. Repitiles
6. Amphibians
7. Molluscs

18. Animals with back bone include:
    A. 1,2,3 and 4
    B. 3,4,5 and 6
    C. 4,5,6 and 7
    D. 1,3,5 and 7
19. Plant groups that do not posses true leaf, stem and root are:
    A. Gymnosperms
    B. Angiosperms
    C. Ferns
    D. Mosses
20. Bats are evolutionarily phyla of sub – kingdom algae are incorrectly matched with their distinctive examples?
    A. Dinoflagellates – Cryptomonas
    B. Bacillariophyta – Diatoms
    C. Chlorophyta – Chlorella
    D. Englenophyta - Chlamydomonas

21. Which of the following phylas of sub – kingdom algae are incorrectly matched with their distinctive examples?
A. Birds C. Sheep
B. Cows D. Goats

22. The animal like characteristics of euglenophytas is:
A. Have flagella for locomotion
B. Have nervous system
C. Autotrophic mode of nutrition
D. Having circulatory system

23. Which one of the following is wrong about fungi?
A. Heterotrophic mode of nutrition
B. Cell wall composed of chitin
C. Digestion takes place inside their body
D. Store carbohydrate in the form of glycogen

24. The mutualistic association between fungus and roots of the plant is:
A. Lichen C. Mycorrhizae
B. Rhizospore D. Parasitism

25. Mushroom is grouped under which phylas of kingdom fungi?
A. Zygomycetes C. Basidomycotes
B. Ascomycetes D. Deletromycotes

26. The smallest green land plants are:
A. Bryophytes C. Gymnosperms
B. Pteridophytes D. Sphenophytes

27. Which of the following are vascular plants?
A. Mosses C. Algaes
B. Ferns D. Liverworts

28. Which of the following is true about gymnosperms? They have:
A. Flower C. vessels in xylem
B. Seeds enclosed in ovary D. needle shaped leaves

29. All of the following are examples of monocots except:
A. Wheat C. Teff
B. Bean D. Maize

30. Apperiance of netted vennation on leaves is the characteristics of
A. Ferns C. Monocots
B. Mosses D. Dicots

31. Phylas of kingdom animalia that contains all vertebrate animals is:
A. Porifera C. Mollusca
B. Chordate D. Arthopoda

32. Organisms
following ar
A. Have
B. They
C. They
D. Have

33. The simple
A. Spo
B. Cni

34. Which of t
A. Fila
B. Asc

35. Among th
contains e
A. Co
B. Ne

36. Tiny hair
A. Fl
B. Ci

37. The mos
phylum.
A. N
B. A

38. Which
A. C
B. A

Match the foll
matches.

39. Porifer
40. Platyhe
41. Nemat
42. Anneli
43. Mollu
44. Echin
45. Arthro

32. Organisms of kingdom animalia are characterized by all of the following are except.
   A. Have eukaryotic cells
   B. They decompose dead bodies
   C. They exhibit locomotion
   D. Have heterotrophic mode of nutrition

33. The simplest of all invertebrates are
   A. Sponges
   B. Cnidaria
   C. round worms
   D. segmented worms

34. Which of the following animal has segmented body?
   A. Filarial
   B. Ascaris
   C. Earthworm
   D. sponges

35. Among the following phylas of kingdom animalia, which one contains evolutionary dead organisms?
   A. Coelentrata
   B. Nematod
   C. Annelida
   D. poriFera

36. Tiny hair like structures that helps annelids for locomotion are:
   A. Flagella
   B. Cilia
   C. setae
   D. pseudopodia

37. The most intelligent groups of all invertebrates are found in the phylum.
   A. Nematoda
   B. Annelida
   C. Mollusca
   D. Arthropoda

38. Which classes of phylum Arthropoda posses no antenna?
   A. Crustecea
   B. Arachinida
   C. Insecta
   D. chilopoda

Match the following phylas of kingdom animalia with their respective matches.

A	B
39. Porifera	a. Soft bodied animals
40. Platyhelmentes	b. Joint – footed animals
41. Nematoda	c. Sponges
42. Annelida	d. Cnidaria
43. Mollusca	e. Round worms and nematodes
44. Echinodermata	f. Spiny – skinned animals
45. Arthropoeda	g. Segmented worms
	h. Flat worms

Fill in the blank space

46. The group of organisms that can interbreed and produce fertile off spring are called _____.
47. _____ is the study of classification of organisms.
48. A limb in a five digit plan is called _____.
49. In a scientific name the first name is _____, and the second name is _____.
50. All prokaryotic organisms are included in the kingdom _____.
51. The representative groups of kingdom monera are _____ and _____.
52. _____is the mutual association between fungus and roots of the plant.
53. The simplest green land plants are _____.
54. The prominent leaf of fern is _____.
55. _____ are called naked seed plants.
56. Animals without back bone are called _____, while animals with back bone are called _____.
57. A body that is divided in to two halves which are the mirror images of one another are _____.
58. Antipblobastic body has layers called _____, _____ and _____.
59. The most numerous groups of all invertebrates are _____.
60. The two body parts of arachnoids are _____ and _____.
61. The excretory organs of insects are _____.
62. _____ is the outer protective structure of crustaceans.
63. The breathing organ of Pisces is called _____.
64. _____ are vertebrates that spend part of their lives in water and part of it on land.
65. _____ are mammals that develop their immature young in their pouch.
66. _____ are called higher mammals.

Answers to Posttest 1

1. C 2. B 3. D 4. C 5. D

Answers to Posttest 2

1. C 2. B 3. D 4. C 5. D

Answers to Posttest

1. C 2.

Answers to Posttes

1. B 2.

Answers to Post T

1. B 2.
6. A 7.

Answer to

1. B 2.
6. C 7.
11. B 12.
16. D 17.
21. D 22.
26. A 27.
31. B 32.
36. C 37

Matching

39. C 40
44. F 45

Fill in the Blac

46. Species
49 Genus n
51. Bacteria
53. Bryophy
56. Inverteb
57. Bilatera
58. Ectoder
59. Nemato
60. Cephal
61. Malphi
64. Amphi

Answers to Posttest 3

1. C 2. D 3. D 4. C 5. A

Answers to Posttest 4

1. B 2. B 3. D 4. C 5. D

Answers to Post Test 5

1. B 2. B 3. D 4. B 5. C
6. A 7. A 8. A

Answer to Supplementary Exercise on Unit Five

1.	B	2.	C	3.	D	4.	A	5.	B
6.	C	7.	C	8.	B	9.	B	10.	D
11.	B	12.	D	13.	A	14.	C	15.	C
16.	D	17.	D	18.	B	19.	D	20.	A
21.	D	22.	A	23.	C	24.	C	25.	C
26.	A	27.	B	28.	D	29.	B	30.	D
31.	B	32.	B	33.	A	34.	C	35.	D
36.	C	37.	C	38.	B				

Matching

39. C 40. H 41. E 42. G 43. A
44. F 45. B

Fill in the Black

46. Species 47. Taxonomy 48. Pentadectayl limb
49 Genus name, species name 50. Protista
51. Bacteria and blue green algae 52. Micorrhizae
53. Bryophytes 54. Frond 55. Gymnosperms
56. Invertebrates, vertebrates
57. Bilaterally symmetric
58. Ectoderm, mesoderm, endoderm
59. Nematodes
60. Cephalothoraxes, abdomen
61. Malphigean tubules 62. Carapace 63. Gill
64. Amphibians 65. Marsupials 66. Placental mammals

Unit – 6

6. Environment

6.1 Ecosystems

Organisms interact with one another and things in their surrounding habitat in order to survive. This means any organism cannot survive unless it interacts with its surrounding environment.

Ecology is a branch of biology which studies about environment.
(Greek word, Oikos = house, - logy = study)
– It studies the interaction of organisms with one another and their environment.

Environment – is the sum of all the living (Biotic) and non – living (abiotic) components a given habitat.

A structure of life and interaction between organisms and their surrounding can be shown in the following relationships based on increasing level of their complexity.

Organism→population→ community→ ecosystems →biome →biosphere

Organism – is an individual living thing or component. It can be animal, plant or microorganisms

Population is a group of organisms of the same species living together in the same habitat.

Examples: human population living in some village.
Eucalyptus tree population found in a given forest habitat.

Community – is a group of organisms of different populations living together in the same habitat. For example a given community may contain different species of organism living together in same habitat, this may include different species of animals, birds, trees, insects, etc.

Ecosystem – involves all organisms in a given habitat and their interaction with their non – living environment. The main non – living components that are vital for all forms of life are water, sunlight, air, etc. Accordingly organisms interact with one another and their non – living environment through different cycles such as: **water cycle, Nitrogen cycle, carbon cycle etc.**

Biomes – are geographical areas which have their own characteristics climatic conditions including **the rainfall, types of vegetation** and **animals**. The major biomes found on different regions of the earth are, **Desert, grasslands, tropical rain forest, temperate forests, tundra, savannah etc,**

Biosphere (Ecosphere) – Is the thin layer of the atmosphere at the surface of the earth, including the biologically inhabited soil, water, air where radiant energy of the sun can be used. It is simply the sum of all the ecosystems on the world.

An ecosystem can be affected by living things (biotic components) and non – living things (abiotic components) of an environment.

i) Abiotic Components

Abiotic components are the non – living factors that can affect the living condition of a given ecosystem. The major abiotic factors include **water, sunlight, air, soil, temperature, humidity, wind, topography of the land, rainfall.**

Water – is one of the most important component in an ecosystem. It serves as a habitat for aquatic organisms, source of drinking water for most animals including land dweling (terrestrial) animals, for irrigation in farming activity, etc.

Sunlight – is the primary source of energy for all forms of life in which it serves as the main source of energy for building organic substances in the process of photosynthesis.

Air – is the mixture of different component gases which are very important. Some of them are oxygen, carbondioxide, nitrogen, etc.

Soil – is the characteristic nature of the soil whether its fertile or not, and its pH level (acidity or alkalinity) can determine the vegetation cover of an ecosystem, which is an important factor in determining the distribution of all forms of life in an ecosystem.

Temperature – determines the ability of an organism to survive within a given ecosystem. This happens because its difficult for

some organisms in an environment with extremely high temperature and difficult for others to survive at low temperatures.

Humidity – is the concentration of water vapor (moisture) in the air. The higher the water vapor in the air, the higher the humidity in the air.
Humidity affects the rate of water loss (transpiration) by the plant. As the air becomes humid, the rate of water lost by the plant through evaporation (transpiration) is less.

Rainfall – is the amount of moisture falling in to an area. The amount of rain falling into an area determines the type and number of plants and animals living there.

Wind – is the current of air moving in the atmosphere. If the air is very windy, its not conducive for some organisms to live in that area. However, wind is important factor in transferring pollen, seeds and fruit in reproduction of some plants.

Topography of the land – Includes altitude, slope and exposure of the land which are important in determining the vegetative cover and suitability of the land for farming activities.

ii) Biotic components

Biotic factors consist of all the organisms living within particular ecosystem. Each organism is the part of another organisms environment, thus they interact in various ways. Some of them are:

Intra-specific factors – occur between members of the same species such as competition for food, territory and finding mate.

Inter-specific factors – those which occur between members of different species. This includes:

- **Predator** – prey relationship which is also called **predation** in which predator animals like lion, tiger feed on other animals (prey) such as cow, rabbit, etc.
- **Symbiosis** – The relationship between two organisms of different species in which one or both get benefited from the relationship. These are.

a) **Commensalism** – is the relationship in which one organism is benefited while the other is neither benefited nor harmed. Example: Relationship between big trees and epiphytes.

b) **Mutualism** – is the relationship in which both organisms get benefited and its an obligatory relationship.
Example: Lichen (algae and Fungi)

c) **Parasitism** – the relationship in which one organism is benefited (the parasite) and another organism is harmed (host).
Example: Relationship between human and parasites like tapeworm.

d) **Protocooperation** – the relationship in which both organisms are benefited but its not an obligatory relationship. Example: The relationship between teeth cleaning bird and crocodile.

Illustrative Examples

1. Which of the following pairs of organisms competition for food is NOT an important factor? (EGSECE, 2004 E.C)
   A. Fish, rat
   B. Sheep, sheep
   C. cow, horse
   D. Hyena, Volture

   Explanation: Sheep and sheep compete on the basis of various things such as food, shelter, finding mate, etc. Their interaction is **intraspecific interaction**.
   - Cow and horse, hyena and voluture compete for food. Cow and horse are herbivores, hyena and vulture are scavengers.
   - There is no need of competition between fish and rat.

   Answer: A

2. Ecology is a biological science that deals with all of the following, except.
   A. Ecosystem
   B. Populations
   C. an organism
   D. community
   E. none

   Answer: E

3. An ecological unit which is self – sufficient in its energy and nutrient requirements is:
   A. Community
   B. Biosphere
   C. ecosystem
   D. population

   Explanation:
   An ecosystem is a self – sustaining, dynamic community of plants and animals in relation to their physical (non-living) environment.

   Answer: C

4. A natural community interacting with its' abiotic environment makes: (EGSECE 1996 E.C)
   A. A population
   B. An organ system
   C. An organism
   D. An ecosystem

Explanation: Ecosystem consists of interaction between organisms of the same and different species with one another and their non-living (abiotic) environment. **Answer: D**

5. In the ocean ecosystem the small fish called remora take the advantage of being transported by sharks. What is the relationship that exists between these organisms?
   A. Protocooporation
   B. Mutualism
   C. commensalism
   D. parasitism

Explanation: Commensalism is a kind of symbiotic relationship between organisms in which one organism is benefited and the other is neither benefited nor harmed. Thus, in the above relationship remora (the small fish) is transported but the shark is neither benefited nor harmed. **Answer: C**

6. Which of the following symbiotic relationships is NOT obligatory?
   A. Mutualism
   B. Protocooperation
   C. commensalism
   D. parasitism

Explanation: Obligatory relationship is a kind of relationship in which the relationship between the two organisms is determinant for their survival. Thus, protocooporation is a kind of relationship between two organisms in which both are benefited but not obligatory. **Answer: B**

7. Any close relationship between two organisms in which one organism lives near, on or even in side another organism and in which at least one organism benefits is known as:
   A. Mutualism
   B. Competition
   C. symbiosis
   D. parasitism

Explanation: Symbiosis is a relationship between two organisms for the benefit of one organism or both.

- ✓ Some popular examples of symbiotic relationship include:
- ✓ **Mutualism** – The relationship between two organisms in which both are benefited and its an obligatory relationship.
- ✓ **Commensalism** – the relationship in which one organisms is benefited, the other is neither benefited nor harmed.

- ✓ **Parasitism** – the relationship in which one organism is benefited **(parasite)** and the other is harmed **(host).**
- ✓ **Protocooporation** – The relationship in which both organisms are benefited but not obligatory.

Answer: C

8. Which of the following refers to the topography of an ecosystem?
   A. Physiographic factor
   B. Climatic factor
   C. Edaphic factor
   D. Biotic factor

Explanation: Topography refers to the physical features of the land such as mountain, hill, slopes, valleys etc.

Answer: A

9. Aquatic autotrophs are limited to a few hundreds of meters of water depth. Which of the following physical factors is directly related to this factor?
   A. Physiographic
   B. Solar radiation
   C. Edaphic
   D. Temperature

Explanation: The depth in ocean to which plants can grow is limited to the area of penetration of sunlight.

Answer: B

10. What differs photosynthetic – bacteria from other photosynthetic authotrophs?
   A. Their source of energy for the production of food.
   B. Their heterotrophic nutrition.
   C. Their source of hydrogen atom to reduce carbondioxide
   D. The type of food they produce in the photosynthetic process

Explanation:

The source of hydrogen for photosynthetic authotrophs like green plants and algaes is **water** (H_2O).

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorphyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2$$

- Photosynthetic bacterias use hydrogen sulphide (H_2S) as a source of hydrogen and they use bacterial chlorophyll which differs from other photosynthetic autotrophs. **Answer: C**

11. The characteristic of soils which have direct or indirect effect on organisms distribution are: (EGSECE 1999 E.C)
   A. Edaphic factors
   B. Climatic factors
   C. biotic factors
   D. physiographic factors

Explanation: Edaphic factor is the characteristic nature of the soil. The characteristic nature of the soil is a determinant factor for the biodiversity of a given area or habitat. **Answer:** A

12. Which relationship best describes the interactions between lettuce and a rabbit? (EGSECE 1998 E.C)
    A. Predator – prey
    B. Parasite – host
    C. decomposer – scavenger
    D. producer – consumer

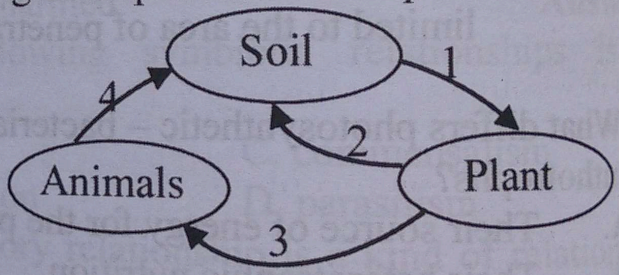
Explanation: In a feeding interrelationship lettuce is a producer and rabbit is a primary consumer. **Answer:** D

13. The physical location of an ecosystem in which a given species lives is called a: (EGSECE 1996 E.C)
    A. Habitat
    B. Community
    C. trophic level
    D. food zone

Explanation: Habitat is a place where an organism lives.

Answer: A

14. Which steps in the diagram represents decomposition?
    A. 2 and 3
    B. 1 and 3
    C. 3 and 4
    D. 2 and 4



Explanation: **Demoposers** – are groups of organisms that digest or break dead particles of an organism. Examples are bacteria and fungus. **Answer:** D

15. Which of the following can be considered as a biotic component in a given ecosystem? (EGSECE, 2006 E.C)
    A. The amount of food available.
    B. The number of predators
    C. The number of parasites and diseases
    D. The type of soils and rocks.

Explanation: Abiotic components are the non – living things that are found in some habitat or ecosystem. This includes water, rock, soil, air, natural gas, minerals, sunlight, etc.

Answer: D

16. List out some examples of biotic and abiotic factors of environment?

Explanation:

- **Biotic components** are the living things in given ecosystem. Thus, examples include; **plants, animals, fungi, bacteria, algae.**
- **Abiotic bacteria, algae** are the non – living things in an ecosystem. These include, **water, light, temperature, soil, air,** etc.

17. Distinguish between

- Intra – specific competition and inter specific competition.

Explanation:

Intra – specific competition	Inter-specific competition
• It is a competition between organisms of the same species. • They compete on the basis of: ➢ Food, territoryfinding mate	• It is a competition between organisms of different species. Examples: ➢ Predation ➢ Symbiosis

Posttest 1

Part I: Choose the best answer from the given alternatives.

1. Inter-specific completion between organisms is on the basis of:
   A. Food C. Predation
   B. Territory D. Finding mate
2. Which of the following is the abiotic component of an ecosystem?
   A. Pant B. Animal C. Water D. Bacteria
3. Which of the following ecological component includes all the rest?
   A. Population C. Ecosystem
   B. Community D. Biosphere
4. The relationship in which the lion hunts and eats zebra is an example of:
   A. Commensalism C. Mutualism
   B. Predation D. Parasitism

5. The win-win and obligatory relationship in which both organisms are benefited from the relationship is:
   A. Mutualism
   B. Parasitism
   C. Protocoporation
   D. Commensalism

6. The biome that gets the highest amount of rain fall per annum is:
   A. Desert
   B. Temperate forest
   C. Grass land
   D. Tropical rain forest

7. In a parasitic relationship between man and tape worm, human being is:
   A. Predator
   B. Prey
   C. Host
   D. Parasite

Part II: Matching

Column A		*Column B*
1 Commensalism	A.	Prey predator relationship
2 Mutualism	B.	Both organisms get benefited but not obligatory
3 Parasitism	C.	One organism benefits and the other is neither benefited nor harmed
4 Predation	D.	One organism benefits and the other gets harmed
5 Protocoportion	E.	Both organisms get benefited and it is obligatory

6.2 Food Relationships

According to their mode of nutrition organisms can be classified as; **autotrophic** and **heterotrophic organisms**.

1) **Autotrophs** – are organisms that can synthesize their own energy from the raw materials in their surrounding environment. Autotrophs can also be classified as:
   a) **Photo autothrophs** – are groups of organisms that use sunlight as primary source of energy to synthesize their organic food materials.
      - These groups of organisms are also called **producers**.
      - **Producers** – are sources of food for other organisms.

      Examples: Green plants, algae, photosynthetic bacteria

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorphyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2\uparrow$$

b) **Chemoautotroph** – are groups of organisms that release energy from simple chemical reactions.
Examples are nitrifying bacteria (nitrosomonas and nitrobacteria)

$2NH_4^+ + 3O_2 \xrightarrow{\text{nitrosomonas}} 2NO_2 + 2H_2O + 4H^+ + \text{energy}$

$2NO_2 + O_2 \xrightarrow{\text{nitrobacteria}} 2NO_3 + \text{energy}$

2) **Heterotrophs** – are organisms that cannot synthesize their own food, rather they feed on other organisms. Since they cannot produce their own food, they are called **consumers**.

- They can be grouped as:
  - **Herbivores**
  - **Omnivores** } macro consumers
  - **Carnivores**
  - **Decomposers** micro consumers

Herbivores – Animals that feed on vegetations (plants).
Examples: Cow, buffalo, giraffe

Omnivores – Animals that feed on both vegetations and animals.
Examples: Human, rat

Carnivores – Animals that feed on animal flesh.
Examples: Lion, cat, tiger

Decomposers – organisms that break dead bodies and decompose them in to the soil.
Example: Bacteria and fungi

Food chain

- It is simple and direct feeding interrelationship which involves transfer of energy and nutrients from one organism to another.
- In food chain energy flow is unidirectional (from producers to consumers).
- Each organism occupies particular **trophic level** within a food chain.

The main trophic levels are

i) **Producers** (First trophic level)

Producers are green plants and algae. They can produce their own energy by the process of photosynthesis.

- Photosynthesis involves conversion of light energy from the sun in chemical energy.
- All organisms depend directly or indirectly on producer.

Note: The sun is the ultimate source of energy for all food chains.

ii) **Primary** (second trophic level) **consumers**

Primary consumers include herbivores that can eat plants directly a omnivores.

iii) **Secondary** (Third prophic level) **consumers**

Secondary consumers are carnivores which mostly eat herbivores a omnivores

iv) **Tertiary** (Fourth trophic level) **consumers**

Tertiary consumers are carnivores (sometimes called **top carnivores** which mostly eat other carnivores.

Additional **quaternary** (fifth trophic level) **consumer** may presen some food chains.

Note: Omnivores can occupy primary and secondary consumer levels.

- **Decomposers** are found at the end of each food chain.

Examples of food chain

1) Oak (leaves) → caterpillar → shrew → owl

Producer	**primary**	**secondary**	**tertiary**
(1st trophicl level)	**consumer** (second trophic) level	**consumers** (third trophic) level	**consumers** (fourth trophic) level

2) Grass → insect → lizard → Birds

3) Phytoplankton → zooplankton → fish → man

Food web

Food web is a complex feeding interrelationship between organisms that consists of many food chains.

➢ Examples of food web

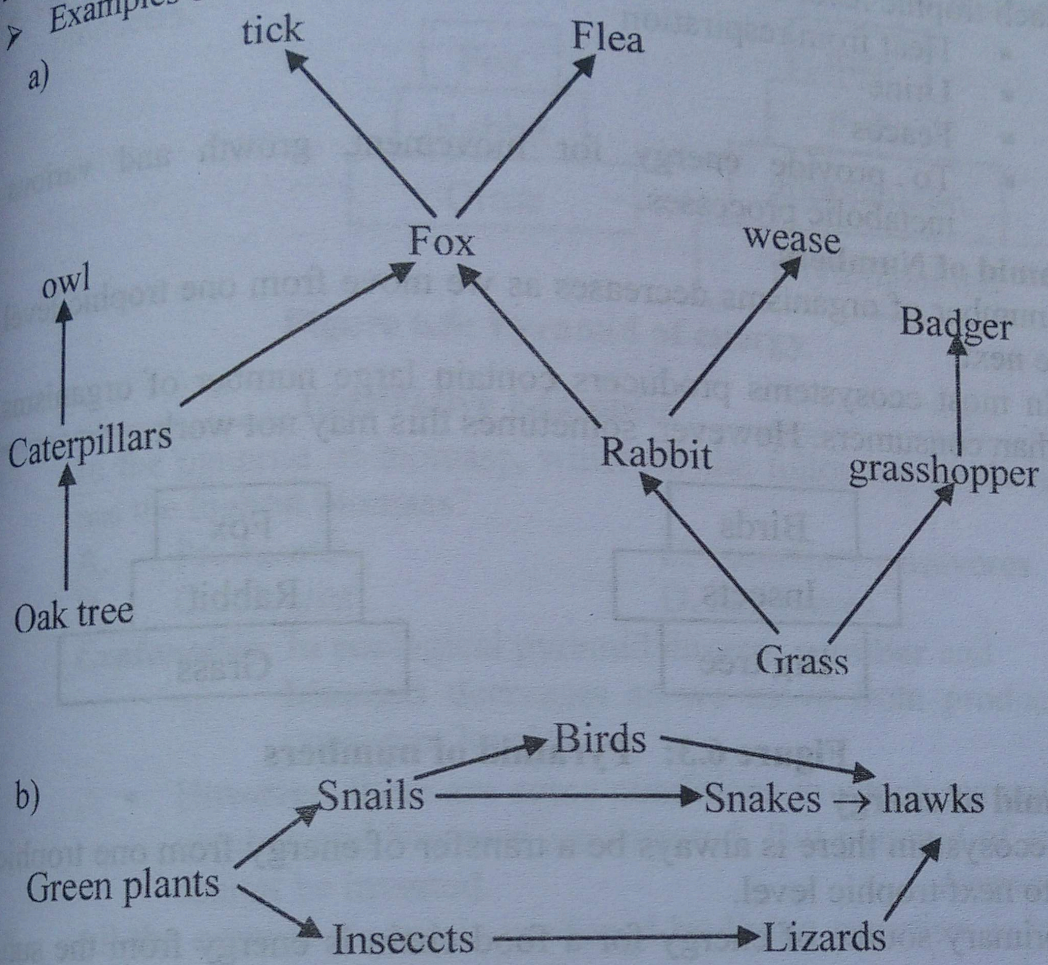


Figure 6.1: Food Web

Pyramid of Biomass

Biomass refers to the total amount of organic matter. The biomass supported at each trophic level decreases as we go from one trophic level to the next. This means the total mass of consumers in particular biomass is less than the total mass of producers.

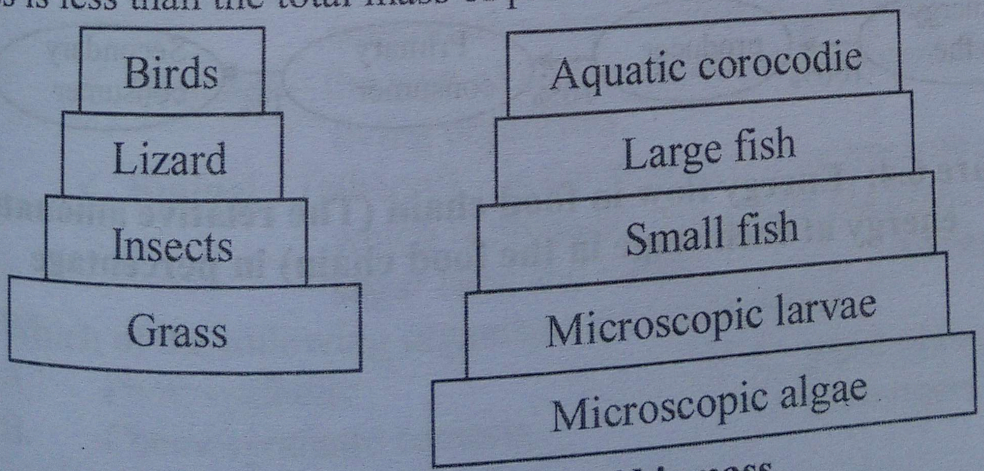


Figure 6.2: Pyramid of biomass

At each trophic level biomass is lost in various forms:

- Heat from respiration
- Urine
- Feaces
- To provide energy for movement, growth and various metabolic processes.

Pyramid of Numbers

The number of organisms decreases as we move from one trophic level to the next.

➢ In most ecosystems producers contain large number of organisms than consumers. However, sometimes this may not work.

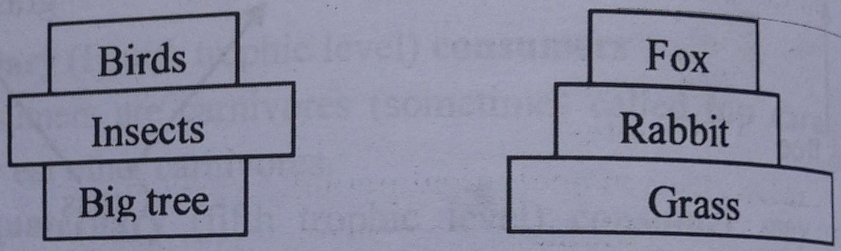


Figure 6.3: Pyramid of numbers

Pyramid of energy

In an ecosystem there is always be a transfer of energy from one trophic level to next trophic level.

The primary source of energy for a food chain is energy from the sun, which is used by producers during photosynthesis. The energy from the sun changes into chemical energy. Then transferred from one organism to another in a food chain.

There is progressively less energy available for organisms further down a food chain.

Energy is cannot be recycled in an ecosystem, it always flows in one direction.

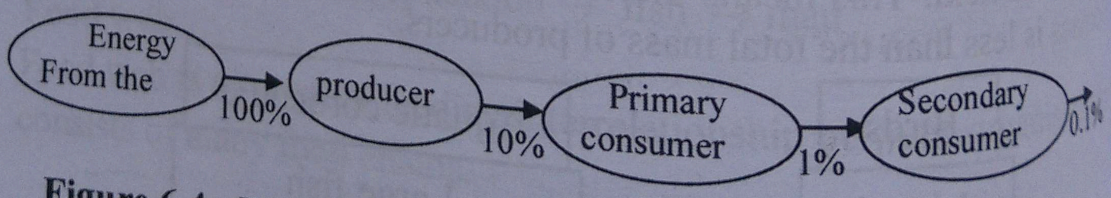


Figure 6.4: Energy flow in food chain (The relative amounts of energy at each stage in the food chain) in percentage

- The figure shows about 90% energy is lost and only 10% is used by producers.

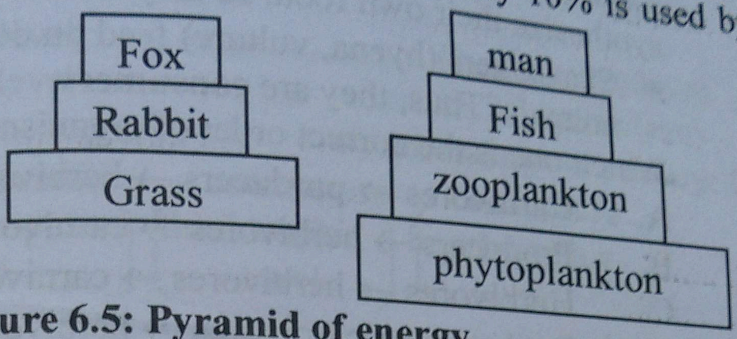


Figure 6.5: Pyramid of energy

Illustrative Examples

17. In the pyramid of biomass, which of the following tropic levels has the highest biomass?
    A. Producers
    B. Carnivores
    C. secondary carnivores
    D. herbivores

    Explanation: In ecological pyramid **energy**, **number** and **biomass** decreases as we move from producer to consumer levels.

    - However, there are some conditions in which pyramids of number and biomass are inverted. But pyramid of energy cannot be inverted.

    Answer: A

18. All the organic materials produced by living organism is known as:
    A. Food web
    B. Food chain
    C. biomass
    D. biofuel

    Explanation:

    ✓ **Food web** - is a complex feeding interrelationship between organisms that consists of many food chains.

    ✓ **Food chain** – is a simple and direct feeding interrelationship between organisms.

    ✓ **Biomass** – the sum of all the organic maters contained by living organisms.

    ✓ **Biofuel** – (ethanol) – is the alternative energy source which is prepared from plant materials like sugar cane and maize.

    Answer: C

19. Which of the following organisms are consumers?
    A. Photosynthetic
    B. Chemosynthetic bacteria
    C. green plants
    D. scavengers

Explanation: Photosynthetic and chemosynthetic organisms can synthesize their own food, so they are not consumers.

- Scavengers (hyena, vulture) feed on dead bodies of plants and animals. Thus, they are **consumers**.

Answer: D

20. Which one is the correct order of organisms in a food chain?
    A. Carnivores → producers → herbivores
    B. Producers → herbivores → carnivores
    C. Herbivores → herbivores → carnivores
    D. Producers → carnivores → herbivores

Explanation: The correct order of feeding relationship between organisms is:

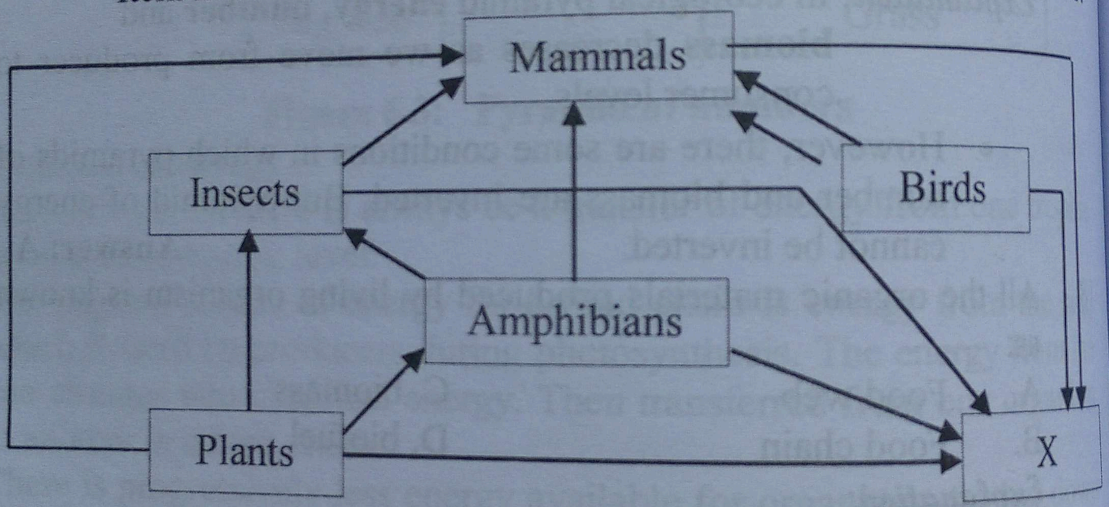
Producers → herbivores → carnivores

Or

omnivores or omnivores

Answer: B

Item number 18 is based on the following diagram of food web.



21. 'x' represents (EGSECE 2004 E.C)
    A. Secondary consumers
    B. Producers
    C. primary consumers
    D. decomposers

Explanation:

Decomposers (bacteria and fungi) break or disintegrate dead bodies of plants and animals, therefore the dead bodies or parts of every organism is decomposed and added in to the soil by decomposers.

Answer: D

22. Which of the following abiotic factors is most important in the water environment?
    A. Type of soil
    B. Amount of rainfall
    C. level of nitrogen
    D. level of oxygen

Answer: C

23. In a simple food chain secondary consumers occupy (EGSECE 2003 E.C)
A. 3rd trophic level
B. 4th trophic level
C. 1st trophic level
D. 2nd tropic level

Producer → primary → secondary → tertiary
(1st trophicl level) **consumer** **consumers** **consumers**
(second trophic level) (third trophic level) (fourth trophic level)

Answer: A

Item 23 is based on the list of organisms given below
Rabbit Hawk Grass snake

24. Which of the listed organisms can be placed in the third trophic level? (EGSECE; 2002 E.C)
A. Rabbit
B. Grass
C. Hawk
D. snake

Explanation:
The correct arrangement of organisms in a food chain is:

Grass → Rabbit → snake → hawk
(1st trophic level) (2nd trophic level) 3rd trophic level) 4th trophic level)

Answer: D

25. Which of the following is a correct food chain in aquatic habitat? (EGSECE, 1999 E.C)
A. Phytoplankton → Fish → zooplankton → tadpole
B. Phytoplankton → zooplankton → tadpole→ Fish
C. tadpole→ Phytoplankton → zooplankton → Fish
D. tadpole→ Fish→ Phytoplankton → zooplankton

Answer: B

Item number 25 is based on the following organisms
1. Scavengers
2. Algae
3. Chemosynthetic bacteria
4. Photosynthetic bacteria
5. Fungi
6. Decomposers

26. Which of the above organisms are autotrophs?
A. 2,3,5
B. 2,3,4
C. 1,5,6
D. 2,4,6

Explanation: Authotrophic organisms are organisms that can synthesize their own energy source. They can be photosynthetic autotrophs (green plants, bacterias and algae) and chemosynthetic bacterias.

Answer: B

27. The reaction, $6CO_2 + 6H_2O \rightarrow C_6H_{12} + 6O_2$, occurs in
A. Chemosynthetic bacteria
B. Photosynthetic bacteria
C. Decomposers
D. algae

Explanation: The reaction above indicates photosynthe process of green plants and algae. Photosynthetic bacteria ha almost related reactions which differs in the source of hydroge The source of hydrogen for photosynthetic bacteria is H (Hydrogen sulphide).

Answer: B

28. Organisms that live and feed on dead organic materials such a decaying wood are known as:
A. Predators
B. Saprotrophs
C. parasites
D. herbivores

Explanation:

- **Predators** – are animals which feed on prey. Examples: Lion, Tiger, etc
- **Saprophytes** – Feed on dead bodies of plants and animals Examples: Hyena, vulture etc.
- **Parasites** – live on or in the body of a living body (host).

Answer: B

29. Which one of the following terms can be applied to all organisms that feed on living and dead organisms?
A. Heterotrophic
B. Autotroph
C. carnivore
D. Herbivore

Explanation:
Heterotrophs are groups of organisms that feed on other organisms since they cannot synthesize their own food.
Examples: decomposers, scavengers, parasites, holozoic organisms

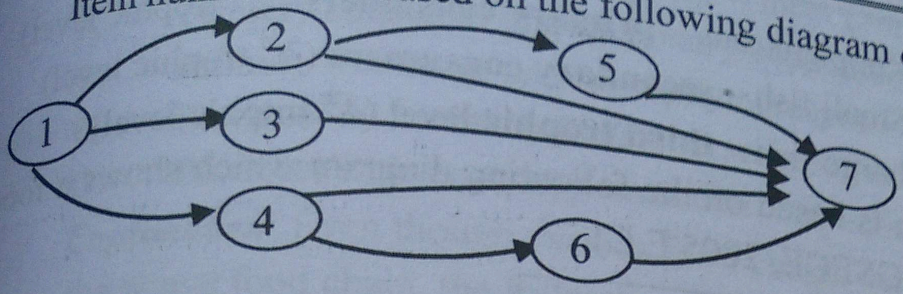
Answer: A

30. Which of the following is true of the pyramid of energy?
A. Primary consumers are large in number than producers.
B. Organisms at higher trophic levels contain more energy than organisms at lower trophic levels.
C. The biomass of carnivores is greater than herbivores.
D. Energy is lost in a food chain at various trophic levels.

Explanation:
In ecological pyramid the amount of energy, biomass and the number of organisms decline as we go down from producers to consumer levels

Answer: D

Item number 30 is based on the following diagram of a food web.



31. Number 4 represents a (EGSECE 2003 E.C)
A. Secondary consumer
B. Producer
C. primary consumer
D. decomposer

Explanation: The level of organisms indicated by each number can be discussed as:

- 1 is a producer.
- 2,3 and 4 are primary consumers.
- 5,6 and 7 are secondary consumers.
- An organism on number 7 can also stand for tertiary consumer.

Answer: C

31. Clearly state the difference between:
i) Photoautotroph and chemoautotroph
ii) Mar consumers and micro consumers

Explanation:

i) Comparison between photoautotroph and chemoautotroph

Photoautotroph	**Chemoautotroph**
• Use sunlight to synthesize their own food. • Have chlorophyll.	Use simple chemical reaction to synthesize their own source of energy. Have no chlorophyll.

ii) **Consumers** are groups of organisms that cannot synthesize their own food and depend on others to find food.

✓ **Macro consumers** are animals which directly feed on others, while **micro consumers** are decomposers.

32. Describe the position of each organism in the following food chain.

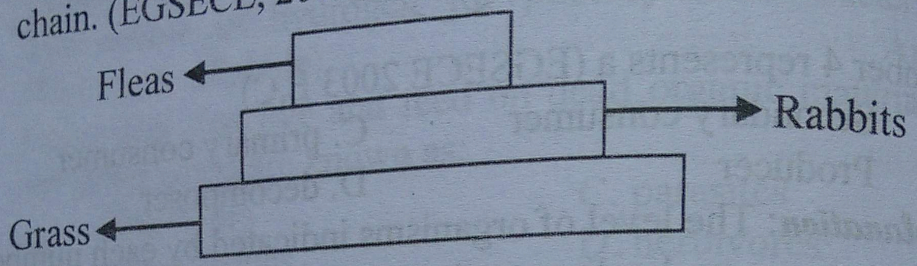
Phytoplankton → zooplankton → small fish → large fish

Explanation:

- Phytoplankton – **producers** (1^{st} torphic level)

- Zooplanktons – **primary consumers** (2nd trophic level)
- Small fish – **secondary consumers** (3rd trophic level)
- Large fish – **third trophic level** (4th trophic level)

33. This item is based on the following diagram which shows a food chain. (EGSECE, 2005 E.C)



Which of the following is true about the above food chain?

A. The number of fleas is greater than rabbits.
B. The number of rabbits is greater than grass
C. Fleas are primary producers
D. Rabbits are primary consumers.

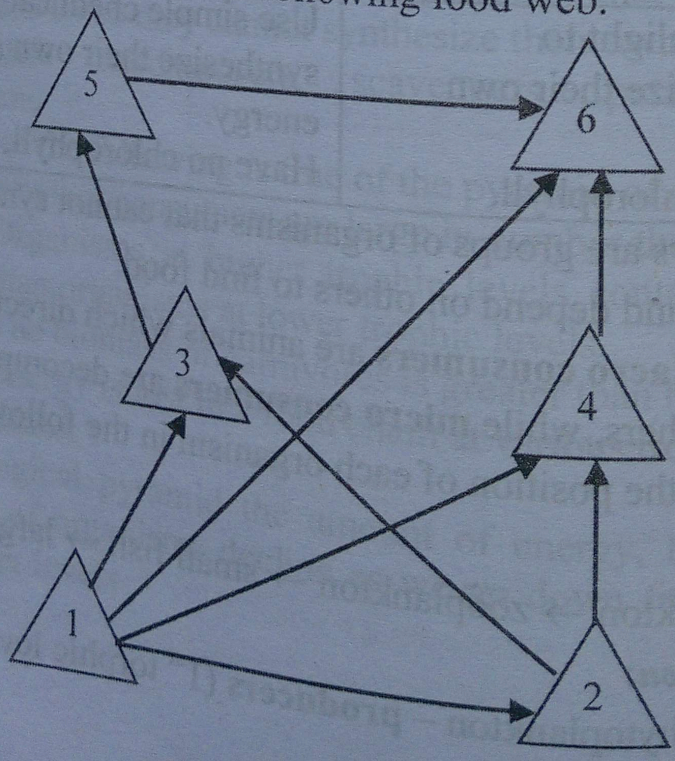
Explanation: According to the above pyramid, the arrangement of organism according to their number is: Grass > Rabbits > Fleas

- In other ways:

Grass ⟶ Rabbit ⟶ Fleas
(Producer) (Primary consumers) (Secondary consumer)

Answer: D

34. This item is based on the following food web.



From the above diagram number 2 and 4 most probably represent

A. Primary consumer and secondary consumer, respectively.
B. Primary consumer and saprophyte, respectively.
C. Producer and primary consumer, respectively.
D. Producer and secondary consumer, respectively.

Explanation: Even though various food chains can be listed from the above food chain, the following food chain can be taken as an example:

1 (producer) → 2 (primary consumer) → 4 (2ndry consumer) → 6 (decomposer)

Answer: A

35. The energy available at each trophic level of food chain is less than it was at previous stage. This is because: (EGSECE, 2006 E.C)

A. The energy available at one stage cannot pass to the above stage.
B. The energy available at one stage is completely passed when it is eaten by above stage.
C. The number of organisms involved at the above stage is greater than the lower stage.
D. Not whole organism at one stage is eaten by the stage above.

Explanation: The amount of energy that is contained at trophic levels decrease as we move from one trophic level to the next. This is because

- Most of the energy is lost at each trophic level during respiration, excretion.
- Not all organisms are eaten and used up by the organisms on the next trophic level.

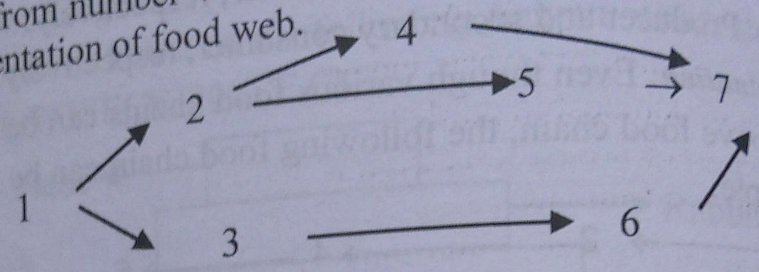
Answer: D

Post Test 2

1. In a food chain an organism that comes first is:
   A. Primary consumer
   B. Producer
   C. Secondary consumer
   D. Carnivores
2. Which of the following groups of organisms are micro consumers?
   A. Carnivores
   B. Decomposers
   C. Herbivores
   D. Omnivores

3. Which of the following organisms are chemo autothrphs?
   A. Nitrobacteria
   B. Photosynthetic bacteria
   C. Algae
   D. Plants

Items from number 4 to 6 are based on the following diagrammatic representation of food web.



4. Herbivores animals are represented by numbers:
   A. 1and 2
   B. 2 and 3
   C. 4,5,and 6
   D. 1 and 7
5. A producer is indicated by number
   A. 1 B. 2 C. 3 D. 7
6. Numbers 4,5,and 6 represent an organism on:
   A. Herbivores
   B. Carnivores
   C. Micro consumers
   D. Producers

Items number 7 to 9 are base on the following food chain

Oak (leaves) → caterpillar → shrew → owl

7. Oak leaves represent
   A. Primary consumer
   B. Secondary consumer
   C. Tertiary consumer
   D. Producer
8. Shrew is _____ in above food chain.
   A. Tertiary consumer
   B. Second trophic level
   C. Third trophic level
   D. Primary consumer
9. A top order consumer in above food chain is:
   A. Oak B. Caterpillar C. Shrew D. Owl
10. Which of the following cannot be recycled in an ecosystem?
    A. Biomass
    B. Energy
    C. Oxygen
    D. Water

3. Recycling in nature

The recycling of substances provides an important link between the abiotic and biotic environment. Substances from the environment are used directly by plants and indirectly by all other organisms. Substances are returned to the environment from organisms by **excretion, egestion** and by **death and decay of tissue.**

Decomposers play a huge role in recycling of nutrients. They are important in various ways to the environment.

1. Break and digest dead bodies of plants, animals and waste materials or droppings of animals and add them into the soil. This increases the fertility of the soil which is useful for plants.
2. Removes unpleasant waste materials from an environment.
3. Making of compost in gardens.

Note: Decomposers are groups of microorganisms that feed on dead bodies of plants and animals.

Examples: Bacteria's and fungi

Recycling of substances in an ecosystem includes **chemical, physical** and **biological** processes.

Examples include **carbon, nitrogen** and **water cycles.**

The nitrogen cycle

Nitrogen is a gas that represents very high proportion of the atmosphere.

- It is important in synthesis of various important biological molecules. This includes **amino acids (proteins)** and nucleic acids such as **DNA** and **RNA** which are responsible for inheritance.
- Nitrogen cannot be directly used by most organisms. Thus the **nitrogen cycle** involves the conversion of nitrogen gas by various biological, chemical and physical processes into usable form by plants.
- The atmospheric nitrogen is fixed by microorganisms living in a symbiotic relationship with certain plants, especially **legumes**. Legumes are plants like pea **(pisum)** and clover **(Trifolium)** have root nodules in which **nitrogen fixing bacteria live** with.
- The **nitrogen cycle** involves the following bacteria.
  1) **Nitrogen fixing bacteria** - convert free atmospheric nitrogen in to nitrate.

     Nitrate – is the only form of nitrogen that can be used by plants.

     Note: Lightning has also a role in fixing atmospheric nitrogen.
  2) **Nitrifying bacteria** – convert ammonium compounds into nitrate.

Ammonium compounds $\longrightarrow$ *Nitrite*

Nitrite $\longrightarrow$ *Nitrate*

3) **Putrefying (amonifying) bacteria's** – They convert the dead bodies and decay of plants and animals in to ammonium compounds

4) **Denitrifying bacteria** – are bacteria's that convert nitrates in to free atmospheric nitrogen.

Plants synthesize protein by using the **nitrates** they absorb from the soil.

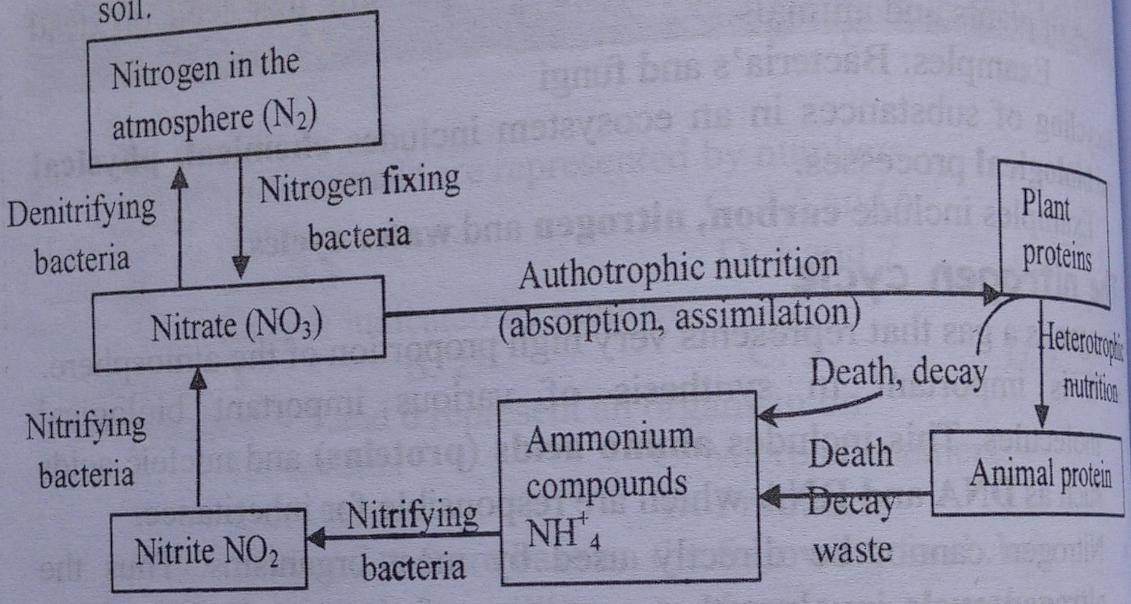


Figure 6.6: The nitrogen cycle

The Carbon Cycle

Carbon is a component of all organic molecules and is essential for all life. The carbon cycle mainly involves the conversion of the inorganic molecule carbondioxide to form organic molecules which are formed within the tissues of organisms.

Note: The major organic substances in living organisms are **carbohydrate, lipid (Fat)** and **protein**.

The concentration of carbondioxide in the air is fairly constant; however there is a possibility of levels gradually increasing because of:

- The increased burning of fossil fuel from cars, factories etc.
- Reduction of vegetations, for instance clearing of forests, which decreases **carbon fixation** by photosynthesis.

An increase in atmospheric carbondioxide would 'trap' retransmitted solar energy which might be lost into space. This is called **green house effect.** This results in global warming which results in increasing earth's average surface temperature.

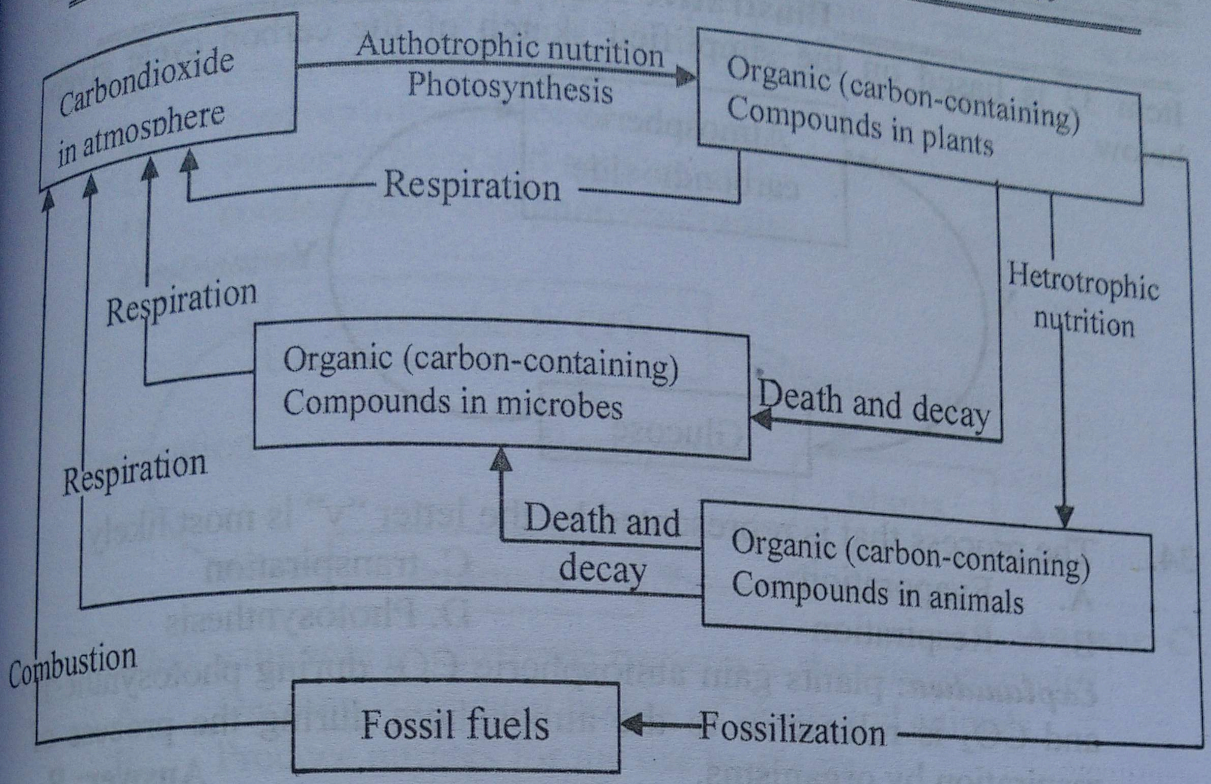


Figure 6.7: The carbon cycle

The Water Cycle

Water is essential for life. Water cycle involves physical process such as **evaporation** and **rainfall**.

- **Transpiration** from plants also contributes to water cycle.

NB. Transpiration is the loss of water from plants by evaporation.

- Rain dissolves some pollutants in the air and these are then deposited on the earth surface on to soil, vegetation and water.
- For example accumulation of sulfuric acid from burning of fossil fuels, which results in **acid rain**.

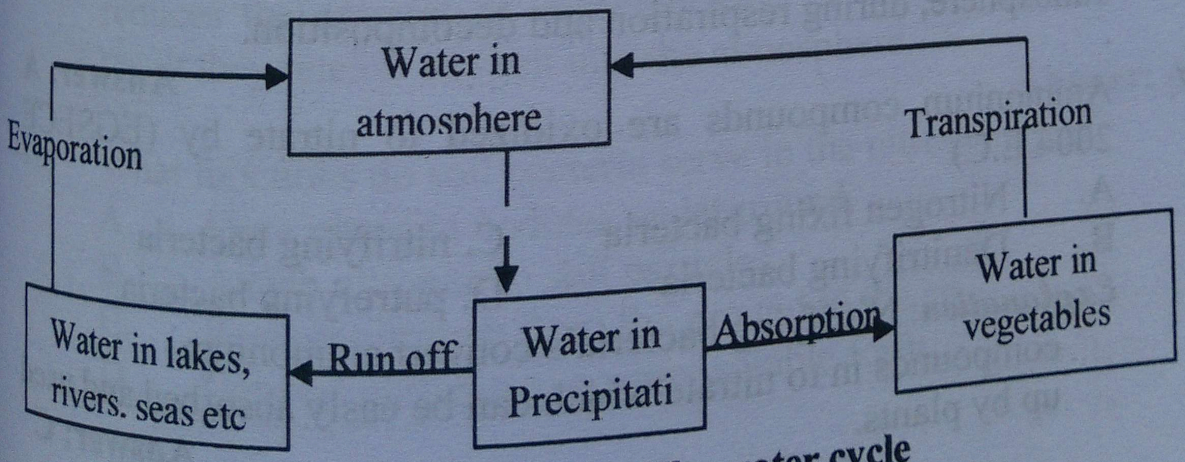
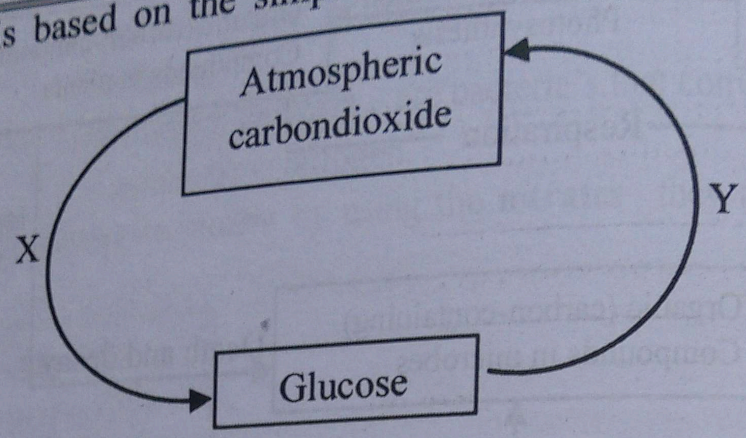


Figure 6.8: The water cycle

Illustrative Examples

Item 33 is based on the simplified sketch of the carbon cycle given below



34. The process that is represented by the letter "y" is most likely
A. Evaporation
B. Respiration
C. transpiration
D. Photosynthesis

Explanation: plants gain atmospheric CO_2 during photosynthesis and CO_2 is release in to the atmosphere during the process of respiration by organisms.

Answer: B

35. Which of the following processes is involved in the carbon cycle?
A. Photosynthesis
B. Condensation
C. transpiration
D. evaporation

Explanation: The carbon cycle involves the two important life processes, namely **photosynthesis** and **respiration**.

Answer: A

36. The pair of processes which return cabondioxide to atmosphere are:
A. Decomposition and respiration
B. Photosynthesis
C. Evaporation and respiration
D. Decomposition and photosynthesis

Explanation: carbondioxide is returned back in to the atmosphere, during respiration and decomposition.

37. Ammonium compounds are oxidized to nitrate by (EGSECE 2004 E.C)
A. Nitrogen fixing bacteria
B. Denitrifying bacteria
C. nitrifying bacteria
D. putrefying bacteria

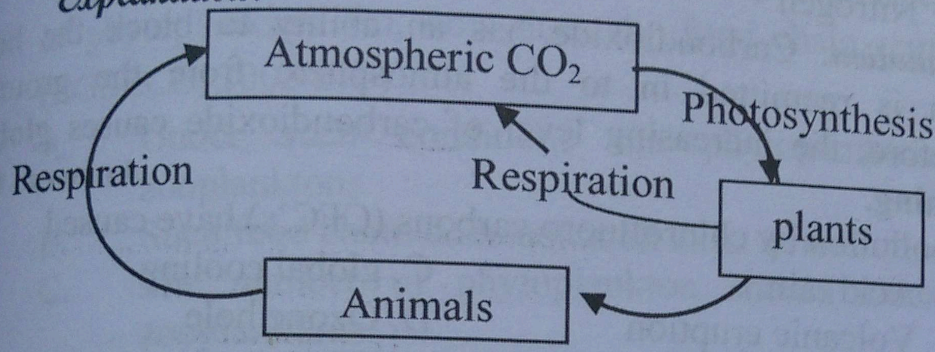
Answer: A

Explanation: Nitrifying bacteria's convert ammonium compounds in to nitrate which can be easly absorbed and used up by plants.

Answer: C

38. Which of the following play an opposite role in recycling carbon between living things and the environment?
   A. Respiration and combustion
   B. Evaporation and condensation
   C. Photosynthesis and respiration.
   D. condensation and photosynthesis

Explanation:



Answer: C

39. What is the role of nitrifying bacteria in the nitrogen cycle?
   A. Decomposing dead bodies of plants and animals.
   B. Produce nitrates for the use by plants.
   C. Release free nitrogen to atmosphere
   D. Build free nitrogen in their bodies.

Explanation: The role of nitrifying bacteria is to convert nitrogen containing substances in to nitrate which can be used up by plants.

Answer: B

40. Which human activity will have the most direct impact on the oxygen – carbondioxide cycle? (EGSECE 1998 E.C)
   A. Destroying large forest areas
   B. Decreasing the use of water
   C. Enforcing laws to prevent the use of loaded gasoline
   D. Reducing the rate of ecological succession.

Explanation: Plants use atmospheric carbondioxide to make their organic food material by the process of photosynthesis which reduces the amount of carbondioxide in the atmosphere. As a result they release oxygen into the atmosphere.

Answer: A

41. What functions do soil bacteria serve in the nitrogen cycle?
   A. Store nitrogen nodules underground.
   B. Attach nitrogen to soil particles
   C. Release nitrogen from the ground to the air
   D. Convert nitrogen to a water soluble form.

Explanation: Nitrogen fixing bacteria's live symbiotically on the root of leguminous plants, which are responsible for attracting (fixing) free atmospheric nitrogen into the soil. **Answer: B**

42. Which of the following gases results in global warming when accumulated in the atmosphere? (EGSECE 2003 E.C)
   A. Hydrogen
   B. Nitrogen
   C. carbondioxide
   D. oxygen

Explanation: Carbondioxide has an ability to block the heat which is reemitted in to the atmosphere from the ground. Therefore, the increasing level of carbondioxide causes **global warming.** **Answer: C**

43. Air pollution by chlorofluoro carbons (CFC's) have caused
   A. Acid rain
   B. Volcanic eruption
   C. global cooling
   D. Ozone hole

Explanation: CloroFluoro carbons (CFC's) are responsible for formation of ozone hole which increases the exposure of the earth's surface to dangerous UV – rays. **Answer: D**

44. Which of the following is correct about the different tropic levels that can exist in a diversified terrestrial ecosystem? (EGSECE, 2005 E.C)
   A. The second trophic level is occupied by herbivores and omnivores.
   B. Herbivores and carnivores occupy the third trohpic level.
   C. The first trophic level is occupied by herbivores and produces.
   D. Carnivores and omnivores occupy the second trophic level

Explanation: In a feeding interrelationship between organisms;
- The fist trophic level is occupied by **producers.**
- The second trophic level is occupied by **herbivores** or **omnivores.**
- The third trophic level is occupied by **carnivores** or **omnivores.**

Answer: A

45. Which of the following biomass is used as energy gain for the organism? (EGSECE, 2005 E.C)
   A. Biomass lost in feaces and urine
   B. Biomass ingested as food
   C. Biomass lost as heat during respiration
   D. Biomass used to provide energy for movement

Explanation:

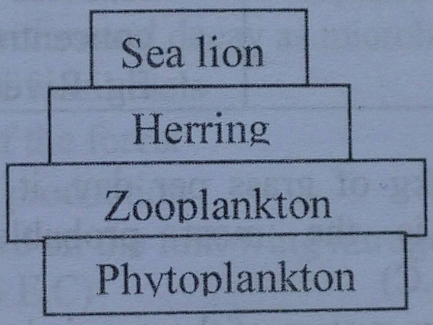
- **Respiration** – is the oxidation of sugar (organic substance) to release energy for the body.
- **Biomass** – is a sum of dry organic matter found in an organism. Therefore, the biomass lost during respiration helps to produce energy gain for the body.

Answer: C

46. In an aquatic ecosystem the biomass of zooplanktons exceeds that of phytoplankton. So, how does the food chain remain stable? (EGSECE, 2005 E.C)

A. Under such conditions the phytoplankton feed on zooplanktons.
B. Such food chain does not at all exist in reality.
C. The number of phytoplankton should exceed that of zooplanktons.
D. The energy contained in phytoplankton is more than that of zooplanktons.

Explanation: for example in some food chains like,



In such kinds of conditions, the energy contained in phytoplankton is more than that of zooplanktons. **Answer: D**

47. Which of the following is true about nitrogen cycle? (EGSECE, 2005 E.C)

A. Pea and beans are legume plants and have nitrogen fixing bacteria.
B. All bacteria in the soil can fix atmospheric nitrogen.
C. All green plants absorb nitrogen gas from the atmosphere.
D. Decomposers break down plant and animal remains and produce nitrogen gas.

Explanation:

- Plants do not directly absorb free atmospheric nitrogen directly. Therefore, they can use nitrogen only in the form of nitrate (NO_3).

- **Nitrogen fixing bacteria** – are found on the root of legume plants like pea and beans to convert free atmospheric nitrogen (N_2) in to nitrate (NO^-_3) in the soil.
- **Ammonifying bacteria** – convert nitrogen containing substances in dead bodies in to ammonium compounds.
- **Nitrifying bacteria** – convert ammonium compounds in to nitrate.
- **Denitrifying bacteria** – convert nitrates in to free atmospheric nitrogen.

Answer: A

48. Which of the following water habitat is marine? (EGSECE, 2005 E.C)
A. Lake
B. Pond
C. Ocean
D. River

Explanation:

Aquatic habitat	
Marine water habitat	***Fresh water habitat***
➢ Has high salt concentration Eg. Sea, ocean	➢ Has less or not slat concentration. Eg. River, pond, stream, lake

Answer: C

49. If a cow eats 10kg of grass per day, it does not increase its biomass. What is the most probable reason for this? (EGSECE, 2005 E.C)
A. The digestive system of the cow is less efficient
B. The grass has little nutrient to be converted to biomass
C. Much of the ingested grass is used as energy source.
D. The cow might have many parasites in its stomach.

Explanation:

- Biomass is a dry organic matter processed by an organism.
- Each organism uses its biomass during respiration as a source of energy. This why biomass decreases as we move from one trophic level to the next.

Answer: C

50. List out some of the bacteria which are involved in nitrogen cycle

Explanation:

✓ **Nitrogen fixing bacteria** – convert atmospheric nitrogen in to nitrate.
✓ **Nitrifying bacteria** – convert ammonium compounds in to nitrate.

- ✓ **Putrefying bacteria** – convert dead bodies in to ammonium compounds.
- ✓ **Denitrifying bacteria** – convert nitrates in to free atmospheric nitrogen.

51. What are the two most important life processes which are involved in carbon cycle?

Explanation: The two most important life processes which are involved in carbon cycle are **photosynthesis** and **respiration**.

52. Which of the following processes occurs during carbon cycle? (EGSECE, 2006 E.C)

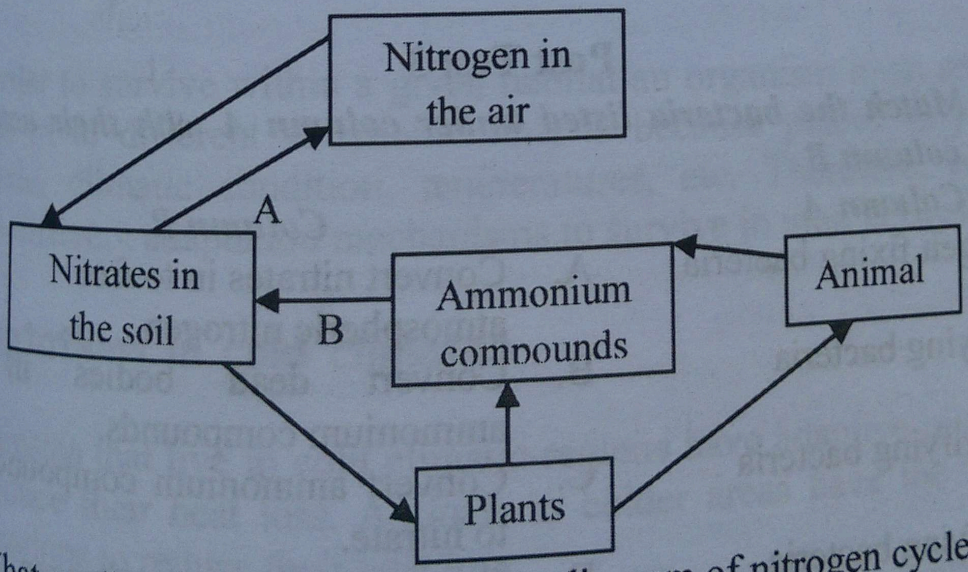
A. CO_2 is removed from the air during burning of fossils.
B. CO_2 is released to the air during decomposition.
C. CO_2 is removed from the air during respiration.
D. CO_2 is released to the air during photosynthesis.

Explanation: The carbondioxide gas is released in to the atmosphere during:

- Respiration from animals and plants.
- Decomposition and decay as microbes respire
- Burning of fossil fuels
- Burning of the forest
- Industries production

Answer: B

53. This item is based on the following nitrogen cycle (EGSECE, 2006 E.C)



What are "A" and "B" on the above diagram of nitrogen cycle?

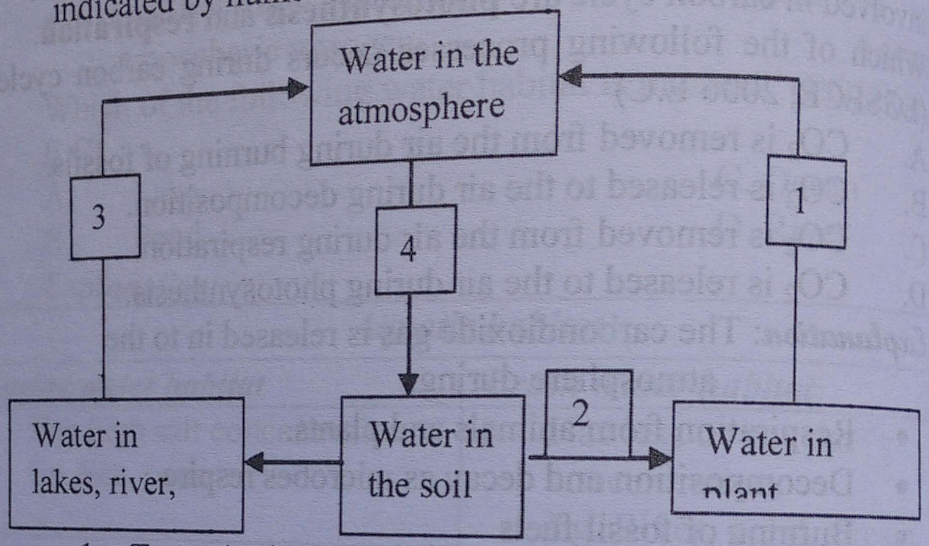
A. Denitrifying and putrefying bacteria, respectively.
B. Nitrifying and putrefying bacteria, respectively,
C. Nitrifying and denitrifying bacteria, respectively.
D. Puterifying and denitrifying bacteria, respectively.

Explanation:

- **Denitrifying bacteria** – convert nitrates in the soil in to nitrogen in the air.
- **Putrefying bacteria** – convert ammonium compounds to nitrates

Answer: A

54. Describe the processes involved in the water cycle below as indicated by numbers.



1 = Transpiration
2 = Absorption
3 = Evaporation
4 = Precipitation (rain, snow)

Post Test 3

Part I: Match the bacteria listed under column A with their actions under column B

Column A	*Column B*
1 Nitrogen fixing bacteria	A. Convert nitrates in to free atmospheric nitrogen.
2 Nitrifying bacteria	B. Convert dead bodies in to ammonium compounds.
3 Denitrifying bacteria	C. Convert ammonium compounds in to nitrate.
4 Putrefying bacteria	D. Convert atmospheric nitrogen in to nitrate.
	E. Convert nitrogen containing substances in dead bodies in to ammonium compounds.

Part II: Choose the best answer from the given alternatives.

1. Which of the following is NOT a fresh water habitat?
   A. Lake B. River C. Stream D. Ocean
2. The two very important processes that are involved in a water cycle are:
   A. Precipitation and respiration
   B. Photosynthesis and respiration
   C. Transpiration and photosynthesis
   D. Transpiration and evaporation
3. Which of the following processes are responsible for returning water in to the atmosphere?
   A. Precipitation and respiration
   B. Photosynthesis and respiration
   C. Transpiration and photosynthesis
   D. Transpiration and evaporation
4. The process by which carbon dioxide in the atmosphere directly changes to carbon dioxide in the atmosphere is:
   A. Respiration C. Photosynthesis
   B. Combustion D. precipitation
5. The usable form of nitrogen by plants is:
   A. Nitrite B. Nitrate C. Ammonia D. Nitrogen gas

6.3 Adaptations

In order to survive within a given habitat an organism must adapt itself to fit in to different situations. This is because places vary in their altitude, climatic condition, temperatures, etc. Therefore, organisms have different adaptation mechanisms to survive in their ecosystems.

Adaptations in cold climates

Organisms that live in cold climatic regions have adaptive mechanisms to reduce their heat loss. Animals in colder areas have the following adaptations to reduce their heat loss.

- Reduced surface area to volume ratio. Organisms with smaller surface area to volume ratio lose heat slowly and also reduced ear size and other parts of the skin layer to reduce surface area through which heat is lost.

- Thick layer of fat under the skin to keep to store more internal heat.
  Note: Fat has high heat storage capacity.
- **Camouflage** – Is a condition in which arctic animals change their skin surface with seasonal changes.
  - This helps an organism to hide from the prey and for predators not to be seen by preys.
  - For example have white skin in winter and grey or brown in summer.

Examples of animals which live in cold climatic region involves polar beer.

Adaptations in dry climates

In order to survive in dry climates (desert) organisms need to have the following adaptations. Adaptations of plants to hot (desert) climate include:

- ✓ They have long roots to absorb more water
- ✓ They have thick and fleshy steam store more water **(succulents)**
- ✓ They have reduced number of stomata on the leaf surface to reduce the amount of water loss (transpiration)
- ✓ Their leaf surface is covered by waxy cuticles which reduce the loss of water.

- Adaptations of animals to survive in hot (desert) climates include:
  - ➢ They have increased surface area to volume ratio to lose more heat.

    Note: Organisms with large surface area to volume ratio lose heat faster.
  - ➢ They get more water from the food which they eat rather than drinking
  - ➢ Animals like camel – also called 'sheep of the desert' have an adaptation to store water in its hump which helps to stay with out drinking water for 20 – 30 days.
  - ➢ Additional extensions like large ear to increase its surface area through which heat is lost.
  - ➢ Some animals hide themselves in the burrows or shades at the day time to hide from the warm climate and search for food at night.

Some adaptations of Plants

- Wind pollinated flowers have light and feathery pollen.
- Animal dispersed plants have seeds which have hocks and spines.

Some Animal Adaptations

- Carnivore's animals have sharp and pointed canniness to tear Flesh.
- Porcupine (ጃርት) has skin surface which is covered with long stiff parts like needles called quills, which it can raise to protect itself when it's attacked by enemy.

6.4 Tree growing Project

As the human population number increases, the unlimited need of resources of human being is also increasing. Accordingly people are using resources (both renewable and non-renewable).
Trees (vegetations) are one of this natural resources which people are using widely without thinking about the fate of the next generation.

Deforestation – the process of cutting trees without planting another tree for replacement.

- For example if you take Ethiopia, out of the Ethiopian land about 40% was covered by vegetation (forest). However, with in less than 100 year the vegetation cover of our country has declined to only 3%, which is a big loss. Only forests in western part of the coffee growing areas are left.

Currently Ethiopian government is working on the mechanisms of restoring our vegetation cover by **a tree growing project**. Now a days Ethiopian vegetation cover is showing a sign of progress.
Steps of planting trees and protection:

- Slap the young tree in the soil
- The young trees need to be cared with extra water at least for one year
- They need to be protected from animals

Illustrative Examples

55. If you are assigned as a group leader of a tree growing project in your locality. What should be your first task to complete the project successfully? (EGSECE, 2005 E.C)
   A. Design a fund raising program for the project
   B. Tell the society the importance of tree planting
   C. Discuss the way how to continue the project with group members.
   D. Start preparation of the land where trees are going to be planted

Explanation: For a tree growing project, the primary thing to do is creating awareness in a society on how to protect and importance of free planting to their environment. **Answer: B**

56. Which of the following procedure of tree planting is correct? (EGSECE, 2006 E.C)
   A. Prepare the soil ⟶ water it ⟶ dig hole ⟶ plant tree ⟶ press soil around ⟶ use stake.
   B. Prepare the soil ⟶ dig hole ⟶ water it ⟶ plant tree ⟶ press soil around ⟶ use take
   C. Prepare the soil ⟶ water it ⟶ dig hole ⟶ use stake ⟶ plant tree ⟶ press soil around.
   D. Prepare the soil ⟶ dig hole ⟶ water it ⟶ plant tree ⟶ use stake ⟶ press soil around.

Explanation: During planting of a tree the correct sequential steps are:
   - Preparation of soil
   - Digging the hole
   - Water the hole
   - Plant the tree
   - Press the soil firmly
   - Use the stake to support the young tree.

Answer: B

57. Which of the following characteristics is considered as adaptive mechanism for the animal to live in the Arctic? (EGSECE, 2006 E.C)
   A. Dark color
   B. Big ears
   C. large body size
   D. thin fur

Explanation: Arctic region is a region of colder climatic zone. Thus, animals living in this zone have the following adaptive mechanisms:

- Have thick layer of fat under their skin.
- Have reduced surface area to volume ratio including reduced ear size.
- Have fur on the skin
- Have dark color

Answer: A

58. Among the following, which one is animal adaptation to polar areas?
   A. Large surface area to volume ratio.
   B. Thick layer of fat under the skin.
   C. Mechanisms to store more water.
   D. Eating the food that contains large amount of food.

Explanation: polar areas are colder climatic areas. Thus, some of the adaptations of animals in to the polar areas include:

- They develop mechanisms to store more fat under their skin.
- They have furry skin.
- They consume large amount of food (calorie).

59. Which of the following adaptation mechanism is NOT used by polar beer? It has:
   A. Ability to change its skin color with seasonal changes.
   B. Hump for storage of water.
   C. Thick adipose tissue under the skin.
   D. None of the above

Explanation: Polar beer is an animal that mostly lives in polar areas. The adaptive mechanisms used by polar beer include:

- Thick layer of fat under its skin to survive the colder climatic condition.
- Camouflage – its ability to change the color of skin for hiding from its enemies.

60. List out some plant and animal adaptations to desert.

Explanation:

Plant adaptation to desert	Animal adaptation to desert
➢ Have reduced number of stomata and leaves. ➢ Have leaves covered with waxy cuticle ➢ Have long roots ➢ Have thick and fleshy stem , etc.	➢ Have increased surface carea to volume ratio to lose more heat ➢ Having special adaptations like 'hump' in camels which store water for a long time.

61. Describe the major steps of tree growing project.

Explanation:

The steps of planting trees in a tree growing project involve.

- Slap the young tree in the soil.
- Watering the young tree
- Protection from damage by animals.

Post Test 4

Part I: Choose the best answer from the given alternatives.

1. Which of the following is NOT adaptation mechanism of animals in hot climate?
   A. hide themselves in the burrows or shades
   B. eating watery or fleshy food
   C. increased surface areas such as large ear
   D. storage of fat in their body
2. Having increased surface area is an adaptation of animal to:
   A. Cold climate C. Arctic region
   B. Hot climate D. To escape from enemy
3. Adaptive mechanism of animals to live in cold climatic zone is:
   A. Big ear C. Fat storage
   B. Thin skin D. Dark color
4. A condition in which arctic animals change their skin surface with seasonal changes to hide from other animals is:
   A. Adaptation C. Hibernation
   B. Camouflage D. Aestivation
5. Which one of the following is an adaptation of plants to desert biome?
   A. Fleshy stem C. Small leaves
   B. Long roots D. reduced number of stomata
   E. None of the above
6. Which of the following is not in line with tree growing project?
   A. Deforestation C. Reforestation
   B. Afforestation D. Cultivation

1. Which of ecosystem
   A. Sun
   B. Rai
2. Which of
   A. A
   B. A
   C. Th
   D. All
3. Which of
   A. Pre
   B. Ro
4. Animals
   A. Ca
   B. Or
5. Which o chain and
   A. It
   B. It
   C. It
   D. Th
   E. N
6. A green
   A. Li
   B. Li
   C. C
   D. W
7. Among on other
   A. L
   B. M
   C. fe

Supplementary Exercise

1. Which of the following is not an abiotic component of an ecosystem?
   A. Sunlight
   B. Rainfall
   C. grass
   D. wind
2. Which of the following statement is true about communities?
   A. A community is a natural environment which includes all organisms that normally live there.
   B. A community of plants are usually consists of many species of plant.
   C. The individuals in a community of plants or animals depend on each other for some of the necessities of life.
   D. All of the above statements are true.
3. Which of the following are abiotic components of an ecosystem?
   A. Predators
   B. Rocks
   C. parasites
   D. Bacteria's
4. Animals that eat plants only are:
   A. Carnivores
   B. Omnivores
   C. producer
   D. herbivores
5. Which of the following statements does not apply to both food chain and food web?
   A. It involves organisms living in a community
   B. It involves both plants and animals
   C. It involves the transfer of energy from one organism to another.
   D. The energy transferred come originally from the sun.
   E. None of the above
6. A green plant can manufacture starch in the presence of
   A. Light, chlorophyll and carbon dioxide
   B. Light and chlorophyll
   C. Chlorophyll and carbon dioxide
   D. Water light carbon dioxide and chlorophyll
7. Among the following which do not photosynthesize and depends on others for their food are:
   A. Lichens
   B. Mosses
   C. ferns
   D. algae
   E. none of the above

8. One of the following is the correct pattern for the flow of energy in nature.
   A. Solar energy → herbivores → green plant → carnivores
   B. Solar energy → herbivores → carnivores → producers
   C. Carnivores energy→ herbivores→green plant→ solar energy
   D. Solar energy → green plant → herbivores → carnivores
9. Which of the following groups occupies first trophic level?
   A. Carnivores
   B. Omnivores
   C. producer
   D. decomposer
10. Which one of the following groups has the highest efficiency of obtaining energy from grasses?
   A. Primary consumer
   B. Secondary consumer
   C. tertiary consumer
   D. First trophic level
11. The factor which would not affect the habitat of a living organism is
   A. Climatic
   B. edaphic
   C. biotic
   D. none of the above
12. which is the highest level of ecological complexity from the alternatives listed below
   A. community
   B. ecosystem
   C. population
   D. species
13. Which of the following biological system is outside the domain of the biological field ecology?
   A. Population
   B. Community
   C. ecosystem
   D. tissue
14. Organisms that get energy from the breakdown of in organic chemicals is_____
   A. Consumer
   B. Hetrotrophs
   C. chemotrophs
   D. decomposer
15. Microscopic organism which eats phytoplankton is:
   A. Zooplankton
   B. Plant plankton
   C. animal plankton
   D. none
16. The organism that is incorrectly paired with its feeding level is:
   A. Grasshopper – primary consumer
   B. Zooplankton – secondary consumer
   C. Grass – producer
   D. Giraffe – primary consumer

17. Plants which
   A. Legu
   B. Moss
18. The bacteri
   down into n
   A. Deni
   B. Nitro
19. Microscopi
   food chain
   A. Zoo
   B. Phyt
20. Which gro
   plants, that
   A. Dec
   B. Sca
21. The group
   A. Her
   B. Car
22. Which of
   interactio
   complexi
   A. Po
   B. Co
   C. Po
   D. Bi
23. A geog
   conditio
   A. H
   B. B
24. Which
   from the
   A. P
   B. P
25. Which
   the type
   ecosyst
   A.
   B.

17. Plants which have the ability of fix nitrogen in the soil is
A. Legumes C. ferns
B. Mosses D. conifer

18. The bacteria that use nitrates as energy source and break them down into nitrogen gas
A. Denitrifying bacteria C. nitrifying bacteria
B. Nitrogen fixing bacteria D.putrefying bacteria

19. Microscopic photosynthetic organism at the beginning of aquatic food chain are:
A. Zooplankton C. animal plankton
B. Phytoplankton D. None

20. Which group of organism breakdown the remains of animal & plants, that return the nutrient to the soil.
A. Decomposer C. carnivores
B. Scavenger D. omnivores

21. The group of organism that occupy at the end of food chain is
A. Herbivores C. decomposer
B. Carnivores D. producers

22. Which of the following pathway shows the correct order of interactions between organisms according its increasing level of complexity?
A. Population → ecosystem → community → Biosphere
B. Community → Population → ecosystem → Biosphere
C. Population → Community → ecosystem → Biosphere
D. Biosphere→ ecosystem → community → Population

23. A geographical area with its own characteristic climatic conditions including rainfall, types of vegetation and animals is
A. Habitat C. Biome
B. Biosphere D. ecosystem

24. Which of the following component of an ecosystem is different from the rest?
A. Predator C. vegetation
B. Prey D. water

25. Which of the following abiotic factor is most determinant about the type and number of organism (both plants and animals) in an ecosystem?
A. Wind C. Topography
B. Rainfall D. consumers

26. Which of the following are different from the rest in their requirements for photosynthesis?
   A. Photosynthetic plants
   B. Photosynthetic algae
   C. Ferns
   D. photosynthetic bacteria

27. Among the following groups of consumers which ones are called micro consumers?
   A. Herbivores
   B. Omnivores
   C. carnivores
   D. decomposers

28. The position which an organism occupies in a food chain is:
   A. Producer level
   B. Consumer level
   C. trophic level
   D. decomposer level

29. If the following groups of organism are arranged in a food chain which organism comes in the second trophic level?

Lizard	grass
Insects	birds

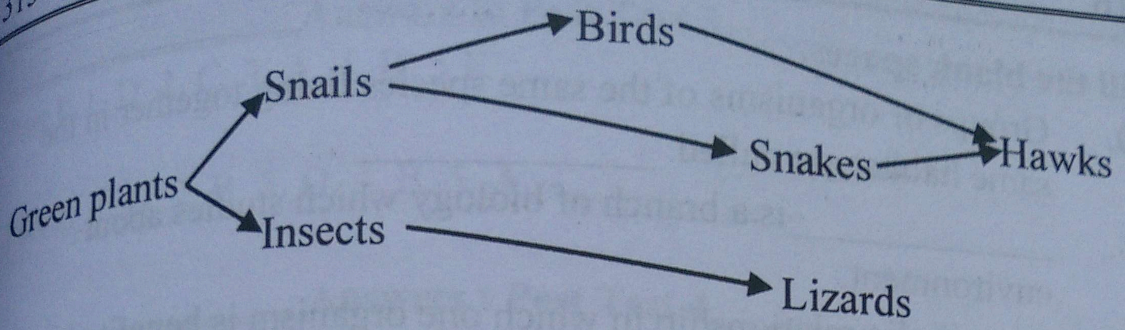
   A. Lizard
   B. Insect
   C. grass
   D. birds

30. Which of the following process does not contribute to the loss of biomass in an ecosystem?
   A. Excreation
   B. Photosynthesis
   C. respiration
   D. egestion

31. An organism that can occupy second and third trophic level is:
   A. Grass
   B. Lion
   C. horse
   D. human

32. In the following aquatic food chain, which organism is incorrectly matched with its relative position?
   Phytoplankton → zooplankton → fish → man
   A. Zooplankton – 2^{nd} tropic level
   B. Phytoplankton – primary consumer
   C. Fish – secondary consumer
   D. Man – 4^{th} tropic level

33. Which of the following cannot be recycled in an ecosystem?
   A. Carbon
   B. Water
   C. energy
   D. phosphorous



Question number 34 is based on the following food web.

34. Organisms on the third tropic level are:
   A. Hawks, lizards, snakes
   B. Birds, snakes, lizards
   C. insects, birds, snakes
   D. hawks, snakes, birds

35. The form of nitrogen that can be directly used by plants is:
   A. Atmospheric nitrogen
   B. Ammonium compounds
   C. nitrite
   D. nitrate

36. Bacteria that convert dead bodies and decay of plants and animals in to ammonium compounds is:
   A. Nitrifying bacteria
   B. Nitrogen fixing bacteria
   C. putrefying bacteria
   D. denitrifying bacteria.

37. Burning of fossil fuels, deforestation and other related activities results in increasing the level of carbon dioxide in the atmosphere, this condition, results in:
   A. Global dimming
   B. Global warming
   C. ozone depletion
   D. all are possible answers

38. Which of the following processes are responsible for the release of water in to the atmosphere
   A. Evaporation and precipitation
   B. Transpiration and absorption
   C. Evaporation and respiration
   D. Precipitation and absorption

39. Which of the following is not adaptation of plants to dry climates?
   A. Long roots
   B. Thick and fleshy stem
   C. Broad leaves
   D. Fewer number of stomata

Fill the blank spaces.

40. Groups of organisms of the same species living together in the same habitat are called _____.
41. _____ is a branch of biology which studies about environment.
42. A symbolic relationship in which one organism is benefited and the other is harmed called _____
43. Organisms that break down the dead bodies of other organisms are called _____.
44. _____ is the complex feeding interrelationship between organisms.
45. The form of nitrogen that can be directly used by plants is called _____.
46. The layer of the atmosphere that protects the earth from harmful ultraviolet rays of the sun is called _____.
47. _____ are features which organisms develop to survive in particular habitats.
48. The ability of an animal to change its colour into its surroundings to hide itself is called _____.
49. Levels in a food chain an organism belong is called _____.
50. _____ are the sum of all the organic material produced by living organisms.

Answers to Post Test 1

Part I: 1. C 2. C 3. D 4. B 5. A 6. D 7. C

Part II: 1. C 2. E 3. D 4. A 5. B

Answers to Post test 2

1. B 2. B 3. A 4. B 5. A
6. B 7. D 8.C 9. D 10. B

Part I: 1.

Part 2: 1

1. D

1. C
2. D
3. B
4. D
5. E
6. D
7. E
8. D
9. C
10. A

Fill in

40.

43.

46.

49.

Answers to Post Test 3

Part I: 1. D 2. C 3. A 4. B 5 E

Part 2: 1. D 2. B 3. D 4. B 5. A

Answers t Post Test 4

1. D 2. B 3. C 4. B 5. E 6. A

Answers to Supplementary Exercises

1. C	11. D	21. C	31. D
2. D	12. B	22. C	32. B
3. B	13. D	23. C	33. C
4. D	14. C	24. D	34. B
5. E	15. A	25. B	35. D
6. D	16. C	26. D	36. C
7. E	17. A	27. D	37. B
8. D	18. A	28. C	38. C
9. C	19. B	29. B	39. C
10. A	20. A	30. B	

Fill in the blank

40. Population
41. Ecology
42. Parasitism
43. Decomposers
44. Food web
45. Nitrate (NO_3)
46. Ozone (O_3)
47. Adaptation
48. Camouflage
49. Trophic level
50. Biomass

Unit – 7

7. Biotechnology

Introduction

The development of biological science has contributed a lot in our daily life activities. The benefits that biology brought to our life can be seen from different areas such as food industries, health, energy production, agricultural activities, and so on. Thus, the knowledge of microbiology (a branch of biology which studies about microorganisms) helped mankind to use microorganisms for human needs.

7.1 What is biotechnology?

Biotechnology is the process in which people use microorganisms to make things that people need. The use of microorganisms has made peoples life easier by solving the practical problems in different area of living. These include:

- ✓ Agriculture
- ✓ Food industries
- ✓ Health
- ✓ Environment

Most people think, biotechnology is quite a modern application. However, looking backwards to what people have been using in their daily activities to produce food, and beverages involves the role of microorganisms.

Example: In Ethiopia

- → Thepreparation of food stuffs such as injera, bread, yoghurt and cheese involves the effect of microorganisms.
- → Preparation of alcoholic drinks such as tej, tella, katikala (ጠጅ) was impossible without the effect of microorganisms.

7.1.1 The Applications of Biotechnology

The biotechnological application can be divided into two major categories. These are:

- ✓ Traditional applications of biotechnology.
- ✓ New applications of biotechnology.

i. Traditional Application of biotechnology

In the traditional application fo biotechnology, the most widely used groups of microorganisms are **bacteria** and **yeast**.

> **DID YOU KNOW?**
> The discovery of microorganisms is not so far as compared to the time where people having been using them for different activities. The first organism was identified by Robert Koch in 18th century. However, in Ethiopia and other parts of the world people have been using yeast (unicellular fungi) for making different types of alcoholic drinks and foods.

Some of the traditional applications of biotechnology are:

- ✓ Production of alcoholic drinks
- ✓ Food production using bacteria

a) Production of alcoholic drinks

The production of alcoholic drinks requires a principal species of yeast known as **saccharomycete**.

- Yeasts are unicellular groups of fungi that can respire aerobically and anaerobically.
- When yeasts respire aerobically they facilitate the baking of bread and Injera.
- When they respire anaerobically in the absence of oxygen they facilitate the production of alcohol (ethanol) from sugar (glucose).

The process by which yeast acts on sugar to produce alcohol (ethanol) is called **alcoholic fermentation**.

The process of alcoholic fermentation is anaerobic.

Note: Anaerobic respiration is the process of respiration that do not utilize oxygen to release energy.

Yeasts actually act on sugar anaerobically to get energy that enables them to carry out their life process called **metabolism**. However, they also produce alcohol (ethanol) as a bi – product from the breakdown of sugar (glucose). The reaction of alcoholic fermentation is indicated as:

$$\underset{\text{(Sugar)}}{C_6H_{12}O_6} \xrightarrow{\text{Yeast}} \underset{\text{ethanol}}{2C_2H_5OH} + 2CO_2 + \text{Energy(2 ATP)}$$

Ethanol (Ethyl alcohol) is a substance that is produced as a result of alcoholic fermentation (anaerobic effect of yeast on sugar). It gives alcoholic drinks and other alcoholic substances to have alcoholic nature.

Alcoholic Products (drinks) in Ethiopia

Traditionally people in our country Ethiopia have a well matured experience of preparing different types of alcoholic drinks. Some of the most commonly known traditional alcoholic drinks in our country are:

- Tej –areke (katikala)
- Tella –borde, etc

Let us see some of them and how they are prepared.

i) Tej

Tej is one of the oldest traditional alcoholic drinks. Preparation of tej requires the following ingredients.

→ **Honey** – consists of sugar that is fermented by yeast.
→ **Water** – to dissolve the fermented sugar (honey).
→ **Gesho** (Rhamnus) sticks – gives tej a bitter taste.

Group work: With your friends in your classroom or study group try to make a discussion about the steps or procedures used to maketej.

ii) Tella

Tella is also another type of traditional alcoholic drink in Ethiopia. It has given different names in different languages of our country. Can you list out some of them? Try it with your peers or study groups.
Tella is prepared from the following ingridents:

- **Cereals** – varied types of cereals are used in the preparation of tella. Some these include; **maize, sorghum, barely**, etc . ..
  - The creals contain sugar that is fermented by yeast to produce ethanol.
- **Bikil** – a germinating seed of cereal. It is used more to produce alchol.
- **Gesho** – to give tella a bitter taste.
- **Water** – the main component needed to dissolve the substances added.

iii) Areke (Katikala)

Katikala is one of the strongest traditional alcoholic drinks in our country. It is prepared by a process of**distillation**.

> Do you know what distillation means? Please refer to your chemistry lesson or ask your chemistry teacher to give you a breaf explanation.

For preparation of Katikala you need to have those substances we used for making of tella with proportionate difference in amount needed.

> **Group task** – Ask people in your locality about the steps and equipment required to make katikala and present it to your pears in the study groups.

Other alcoholic substances that are produced traditionally

iv) Wine

Wine is another traditionally known alcoholic drink that is prepared in different parts of the world as indicated on different historic books.
Wine is prepared from natural sugar that is found in fruits of grapes.
The steps to be followed in the preparation of wine are:

1) Press the fruit and mix the juice with yeast and water.
2) Let the yeast respire anerbically until most of the sugar has been used up.
3) Filter the wine to remove yeast and put it in the bottles, where it matures.

- The wine can be made from almost any fruit or vegetable.
- As the alcoholic content of the yeast gets higher, it kills the yeast and stops fermentation.

b) Production of Food using bacteria

Bacteria are groups of microorganism that belong to the kingdom monera. They are groups of bacteria that are:

- ✓ **Harmful** – causes of different types of diseases.
- ✓ **Useful** – plays role in facilitation of human life.
- ✓ **Neither harmful or useful** – groups of bacteria that do not have both harmful and useful aspect to humans.

- Among the uses of bacteria, they are usedin the production food stuffs such as **yoghurt** and **cheese**.
- Yoghurt and cheese are the diary products produced from milk by the action of the bacterial species known as:
  - ✓ *Lactobacillus bulgaricus*
  - ✓ *Strephtococcusthermophillus*
- The milk consists of a sugar called **lactose**.

Lactose is a disaccharide sugar (carbonydrate) that is found in milk. It is also called**milk sugar**. It is composed of two sugar units known as **glucose** and **galactose**.

Making of Yoghurt

- ✓ The groups of bacteria that are mentioned above act on a lactose (milk sugar) and produce another substance known as **lactic acid**. The process is known as **lactic fermentation**.
- ✓ The lactic acid produced causes the milk to clot and solidify in to **yoghurt**. The action of bacteria also gives yoghurt a smooth and thick texture.
- ✓ Once exist in milk, these useful bacteria prevents the yoghurt from other harmful bacteria's that may cause the milk to get bad or harmful.

Illustrative Examples

1. Biotechnology deals with:
   - A. The structure of micro organisms.
   - B. Diseases and their causes.
   - C. Classification of plants.
   - D. Methods to improve crop yield and quality.

   Explanation: **Biotechnology** deals with the use of micro organisms to make things that people need. Some of these are:

- ✓ Production of alternative sources of energy.
- ✓ Making of different types of drugs and vaccines.
- ✓ Improve crop yield and quality in agriculture.
- ✓ Production of food and alcoholic drinks, etc

Answer: D

2. Which of the following are the waste products of anaerobic respiration by yeast cells?
   A. Ethanol and Carbon dioxides.
   B. Water and lactic acid.
   C. Lactic acid and ethanol.
   D. Carbon dioxide and water.

Explanation: The anaerobic respiration of yeast cells produces the following substances from the anaerobic break down of sugar.

- ✓ Ethanol
- ✓ Carbon di oxide
- ✓ Energy (2ATP)

$$C_6H_{12}O_6 \xrightarrow{\text{Yeast}} 2C_2H_5OH + 2CO_2 + \text{Energy}$$

Answer: A

3. This item is based on the following table which shows different alcoholic drinks and the substances on which yeast cells act.

No.	Alcoholic drinks	Substances on which yeast cells act
1	Tella	Maize
2	Tej	Gesho
3	Wine	Grape
4	Beer	Barley

Which of the alcoholic drinks and the substances on which they yeast act is NOT correctly matched?
A. 2 B. 1 C. 4 D. 3

Explanation:
During the preparation of tej (traditional alcoholic drink in Ethiopia) the yeast acts on a natural sugar found in honey to produce alcohol. Gecho is simply used to give tej a bitter taste.

Answer: A

4. Which of the following activities is the importance of biotechnology?
   A. Mobiles C. Beer
   B. Shoes D. Plastics

Explanation: Beer is an alcoholic drink that is produced by the anaerobic effect of a microorganism called yeast on sugar. **Answer: C**

5. Which of the following an area of biotechnology application production of:
   A. Medicines
   B. Computers
   C. Soft drinks
   D. Plastic bottles

Explanation: Biotechnology plays a significant role in the field of medicine by producing different types of medicines and vaccines. However, making of computers, soft drinks and plastic bottles are not related to biotechnology. **Answer: A**

6. What is the significance of biotechnology?

Explanation: Biotechnology plays a significant role in simplifying our life. These significances are due to the effect of microorganisms such as bacteria and yeast on different substances. The major area in which bio technology is significant are:

- ✓ Energy
- ✓ Food production
- ✓ Health
- ✓ Agriculture etc

7. When a mother wants to make injera, first she prepares dough from teff powder and water. She leaves the dough for two or three days for:
   A. Bacteria to produce water.
   B. Bacteria to produce lactic acid.
   C. Yeast to produce ethanol.
   D. Yeast to produce carbon dioxide.

Explanation: The bubbles of carbon dioxide to make the dough to rise and mix. **Answer: D**

8. Biotechnology is important to:
   A. Produce only alcoholic drinks.
   B. Produce new medicine.
   C. Reduce genetic engineering products.
   D. Reduce new forms of fuel production.

Explanation: Biotechnology is important to:

- ✓ Produce alcoholic drinks, food, and medicine.
- ✓ Increase the quality and yield of agricultural products.
- ✓ Increase genetic engineering products.
- ✓ Introduce new forms of energy such as biogas and bio fuel.

Answer: B

9. Which of the following statement is correct about the use of yeasts? They are used for:
   A. Prevention against disease.
   B. Production of minerals.
   C. Decomposition of dead organisms.
   D. Making food by photosynthesis.

   Explanation: Yeasts are used in:
   - ✓ Production of foods such as bread and injera.
   - ✓ Production of alcoholic drinks.
   - ✓ Decomposition of dead organisms.

   Answer: C

10. List out some examples of alcoholic drinks that are produced traditionally in Ethiopia.

    Explanation: Some examples of alcoholic drinks that are traditionally produced in Ethiopia are:
    - Tej
    - Tella
    - Katikala
    - Borde

Post Test 1

Part I: Fill in the blank spaces provided by using the correct answer.

1. Microorganisms that are widely used in the productions of alcohol and food stuffs such as injera are_____.
2. Biotechnology is the use of _____ to make things that people want.
3. The process by which yeast acts on sugar to produce alcohol is called _____.
4. Yeast acts on a sugar in the absence of oxygen. Such a process that takes place in the absence of oxygen is called_____.
5. Tej is a traditional alcoholic drink which requires _____, _____ and _____ as ingredients.

II. Match the following traditional alcoholic drinks with their source of sugar on which a yeast acts.

Column 'A'	Column 'B'
1. Tej	A. Barley
2. Tella	B. Grapes
3. Wine	C. Honey

Critical Thinking
Traditionally in our country and different parts of the world, people have been making alcoholic drinks. Thus, we have learned that, the process requires utilization microorganisms (yeast). So do you think that those people had an understanding of the whole scientific process that we have learned?

7.2 The New Applications of Biotechnology

In line with the development of biological science, several benefits gained from the use of microorganisms are gained. These applications are called **biotechnology**.

Even though, people have been implementing biotechnology in a traditional way, the new (modern) application involves:

→ Large scale industrial production.
→ Use of modern technologies.

The new application of biotechnology involves the following major areas:

→ Agriculture
→ Food
→ Medicine
→ Energy

The new application of biotechnology involves one of the most modern technologies called **genetic engineering**.

Genetic engineering – is a technique of changing/altering organisms' characteristics (genotypes) by inserting another organisms gene into its DNA.

Genetic engineering helps to give an organism to have desired characteristics. This helps to increase productivity in areas such as **agriculture** and **medicine**.

Genetic engineering has various advantages over the existing technique of breeding by:

→ It always gives desired characteristics.

→ It requires a little space and time.

i) Application of biotechnology in Agriculture

In agriculture, biotechnology is widely used in **genetic engineering, tissue culture** and **development of microbial insecticides.**

a) Genetic engineering in agriculture

Traditionally formers have been using selective breeding in order to develop new varieties of plants and animals. However, this technique has limitations as compared to genetic engineering. Some of these limitations are:

- It do not always give a desired characteristics.
- It takes a lot of time.
- It requires large space.

Genetic engineering overcomes all the limitations of selective breeding.

→ It always gives desired characteristics.

→ It takes a short period of time.

→ It requires a few spaces.

Genetic engineering has contributed the following benefits into agriculture:

- Improvements on growth of plants and animals.
- Improvement on the food value of crops.
- Developments plants that is resistant to drought, disease and pollutants as well as salt tolerant.
- To produce plants that makes their own pesticide chemicals.

Genetic engineering has its own draw backs:

→ Pesticide resistant insects may appear as they eat pesticide forming plants.

→ Genes from genetically engineered plants and animals may spread to other forms of wild life.

→ Genetically modified plants (crops) may not be fertile which makes farmers to buy seeds each year.

b) Microbial insecticides in agriculture

Microbial insecticides are produced from microorganisms (bacteria) that can produce toxics to control insects that destroy crops.

For example, Bacillus bacterium has been used to produce protein toxin against insects such as flies and mosquitoes.

ii) Applications of biotechnology in Food

Biotechnological applications play a significant role in the production of food. Among these:

→ Production of enzymes from genetically engineered bacteria. These enzymes are used:
- In production of processed foods and drinks.
- To clarify beer.
- To convert starch into glucose.
- To break down food used in production of commercial baby food.

→ Production of Fungal protein called **mycoprotein**. These proteins can be incorporated as Foods. Advantages of using Fungal proteins are:
- They have low fat content.
- They are preferred by vegetarians and people who want to reduce the fat in their diet.

Note: Vegetarian are people who do not eat meat and sometimes animal products too.

iii) Application of Biotechnology in Medicine

- Biotechnology is playing a huge role in modern medicine. It is widely used in the production of **vaccines** and **antibiotic**.
- **Penicillin** is the first model antibiotic which is produced from a fungal species called ***Penicillium notation***. It was introduced by an English bacteriologist **Sir. Alexander Fleming** in 1928.

Penicillinis used to treat bacterial infection.

- The production of penicillin in a large scale was made possible in 1945.

Drugs and genetically

Example: By using produce h produced

Genetic Engineering in Medicine

Drugs and hormones that used as medicine can be produced from genetically engineered bacteria's.

Example: Production of insulin from genetically engineered bacteria. By using a technique of genetic engineering it is made possible to produce human insulin from genetically engineered bacteria. The insulin produced is used for people with a problem of diabetes.

- **Insulin** is a human hormone produced by pancreas. It is used to regulate the blood sugar level.
- **Diabetes** is a non – infective disease that can occur due to several reasons. These are two major types of diabetes; **diabetes I and Diabetes II**.

Diabetes I occurs when pancreas fails to produce the insulin hormone.

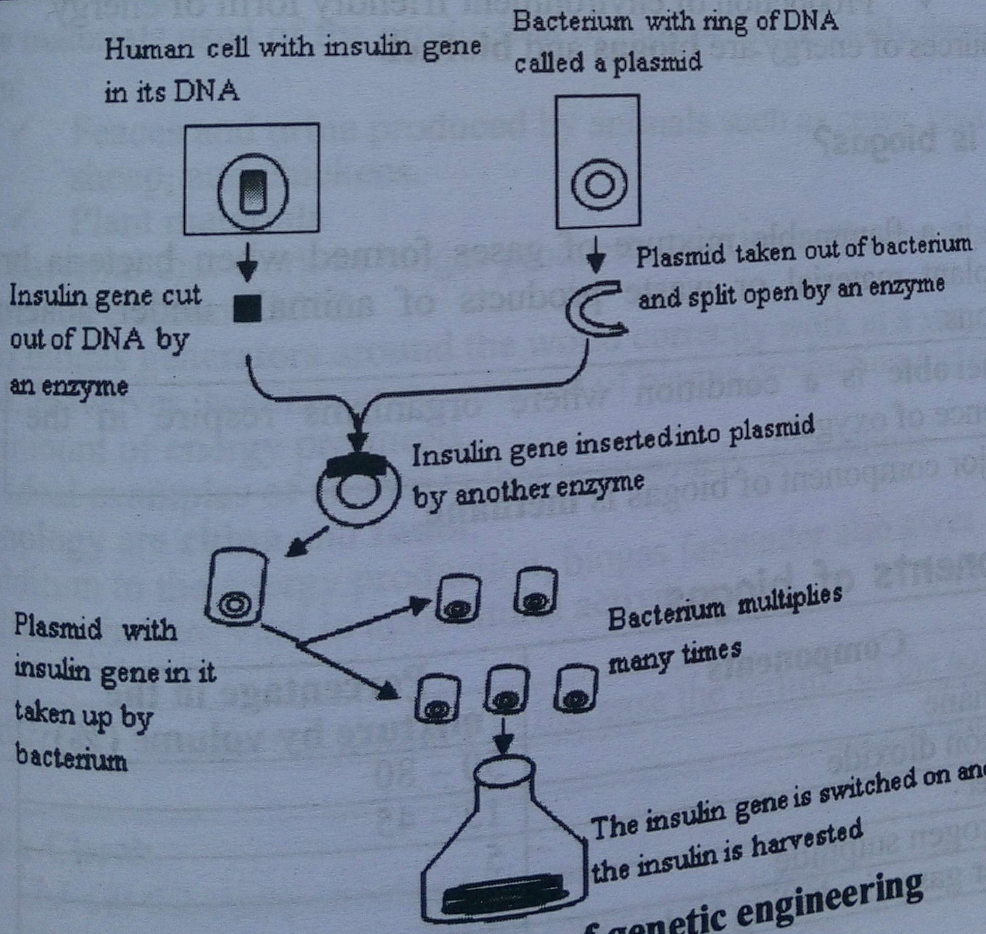


Figure 7.1: The process of genetic engineering

Genetically engineered sheep and other mammals are used to produce lifesaving human proteins. This involves the production of **blood clothing proteins** for example with **hemophilia**.

> **Hemophilia** is a disorder which results in un ability of the body to clot blood after cut or injury.

Other biotechnological applications in the field of medicine are:

- ✓ **Hybridoma** – antibodies that are produced outside the body in hybrid cell cultures.
- ✓ **Interferon** – are proteins that make the body less suspected to be attacked by viruses.

Applications of Biology in Energy Production

Biotechnology has a role in the production of energy sources that are used in cooking, making electricity, fuels for car engines, etc.
There are alternatives sources of energy that are produced by using biotechnology are important in:

- ✓ Using most renewable forms of energy sources.
- ✓ Production of environment friendly form of energy.

This sources of energy are **biogas** and **biofuel**.

What is biogas?

Biogas is a flammable mixture of gases formed when bacteria break down plant material or waste products of animals under anaerobic conditions.

> **Anaerobic** is a condition where organisms respire in the absence of oxygen.

The major component of biogas is **methane**.

Components of biogas

Components	Percentage in the mixture by volume (%)
Methane	50 – 80
Carbon dioxide	15 – 45
Water	5
Hydrogen sulphide	0 – 3
Other gases including hydrogen	0 – 1

Biogas generator/digester – takes in waste materials of plants and animals, it produces **biogas** and **fertilizers**.

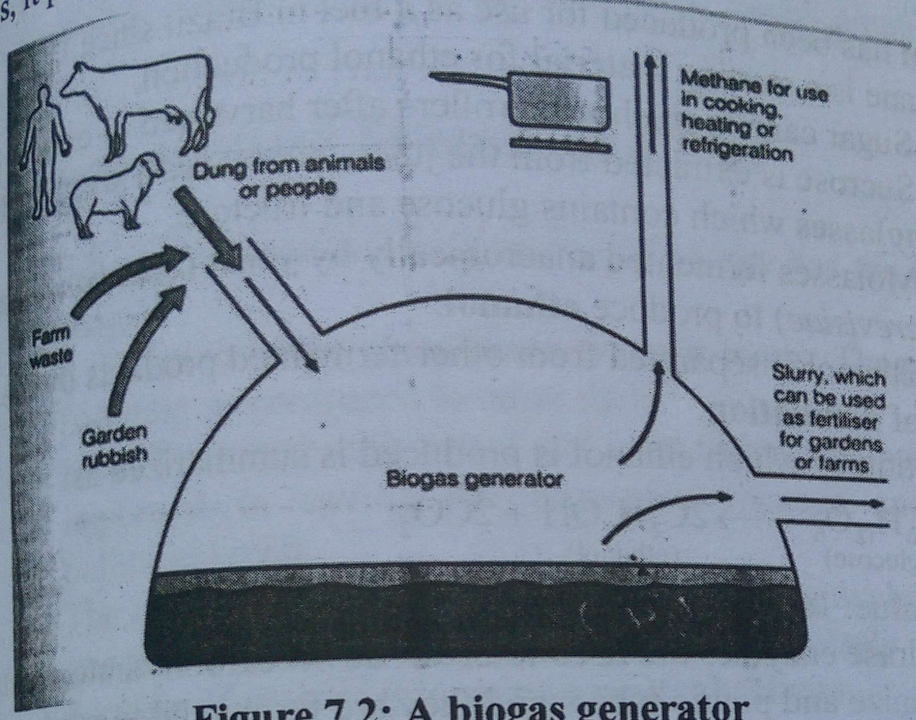


Figure 7.2: A biogas generator

The production of biogas is an **exothermic reaction** which releases or produces energy or heat.

Raw materials used of the production of biogas can be easily found from:

- ✓ Feaces and urine produced by animals such as cows, pigs, sheep, and chickens.
- ✓ Plant materials

The bacterial that work in biogas production are best effective at a temperature around 30°C.

Most biogas generators around the world correctly work at a very small scale. However, scaling up the process of biogas production increases the amount of energy produced.

The ideal examples of countries that have widely used biogas technology are **china** and **India**.

In addition to the energy production, biogas fermenter also gives fertilizers that are vital in agricultural activities.

Fertilizers – are chemicals used to increase the fertility of the soil.

Quick Check

1. List out the components of biogas.
2. What are the raw materials for biogas productions?

Biofuel (Ethanol)

Ethanol has been produced for use as a fuel in Brazil since 1975. Sugar cane is a starting material for ethanol production.

Step 1: Sugar cane is crushed by rollers after harvested to extract juice.

Step 2: Sucrose is extracted from the juice, but leaves a waste called **molasses** which contains glucose and fructose.

Step 3: Molasses fermented anaerobically by **yeast** (*Saccharomyces cerevisiae*) to produce **ethanol**.

Step 4: Ethanol is separated from other fermented products by the process of **distillation**.

The reaction by which ethanol is produced is summarized as:

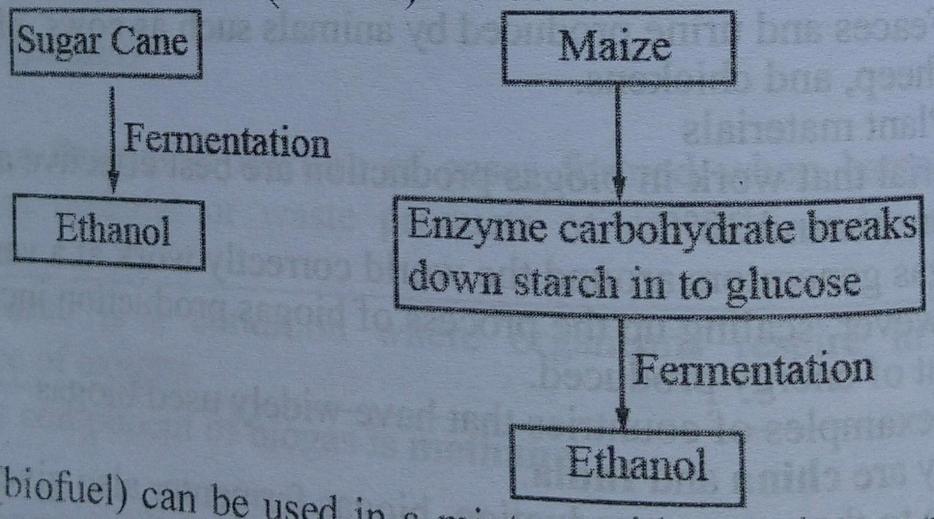
$$\underset{\text{(glucose)}}{C_6H_{12}O_6} \longrightarrow \underset{\text{(ethanol)}}{2C_2H_5OH} + 2CO_2$$

Energy value: 16 kJ g^{-1} 30 kJg^{-1}

Carbohydrase enzymes are used to break down carbohydrate substance such as maize and potato tubers and they are fermented to produce **ethanol**.

> **Carbohydrase enzymes** are types of enzymes that are involved in break down (catalysis) of carbohydrae compounds.

The production of biofuel (ethanol) can be summarized as:



Ethanol (biofuel) can be used in a mixture with petroleum and used to the engines of vehicles including car. The mixture of ethanol and petroleum is called **gasohol**. The proportion of ethanol and petroleum varies.

The mixture of ethanol with petrol is important for countries like Ethiopia in reducing the foreign currency invested on petroleum. Ethiopia has started working on the production of gasoho. There are also countries that have best experience in the production of gasohol.

One of these countries is Brazil and have reduced oil (petroleum) consumption price by 20%. Therefore, the mixing of Ethanol with petrol has a promising impact in the future economic development of our country.

Advantages of Using Ethanol as a Fuel

Utilization of ethanol (biofuel) as a source of energy has the following major advantages:

- ✓ It do not produce toxic gases when it burns. Thus, it is less pollutant as compared to other fuels.
- ✓ It can be mixed with petrol and used for car engines which is important in saving the economic wastage and reduces pollution levels.
- ✓ The carbon that is released from ethanol is equivalent with the one that is released during respiration from plants. Therefore, burning of ethanol is a **carbon neutral** activity.

Disadvantages of Using Ethanol as a Fuel

Even though ethanol has several advantages while used as a source of energy, it has also some disadvantages. These are:

- ✓ It takes a lot of plant material.
- ✓ It requires large space and suitable climate to grow the plants.

Finally, Ethiopia is one of the countries which are suitable for the production of biodiesel. This is because has enough space and suitable climatic condition for growing plants used as raw materials for the production of biofuels (ethanol).

Illustrative Example

11. What is the limitation to use ethanol as a fuel for cars?
    - A. It produces much organic wastes in the environment.
    - B. It emits much toxic gases than conventional fuels.
    - C. It takes a lot of plant materials to produce.
    - D. It increases the level of carbon dioxide in the atmosphere.

Explanation: Ethanol (biofuel) has some limitations. They are:

- ✓ It takes a lot of plant material.
- ✓ It requires large space and suitable climate to grow the plant.

Answer: C